Code report: statistical analysis of MEA synchronization data

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Introduction

This report gives details of the statistical analysis performed , and the code used to generate it, for the manuscript "OPPOSITE MOVEMENT IN FEMALE CONTROLS CHARACTERIZE FIRST ENCOUNTER INTERVIEWS VERSUS FEMALE BORDERLINE PERSONALITY DISORDER THERAPEUTIC INTERVIEWS" by Muñoz-Delgado et al. The aim of this study was to investigate the behavioral synchronization in a clinician-patient dyad versus a control group during a therapeutic/interview session. The following questions were considered:

1) Magnitude of the synchronization (patient following therapist):
2) Which subject, patient or therapist, leads the movement with higher frequency.
3) The sign of the correlation (movement of patient follows or is inhibited by the movement of the therapist).

The database

The database *MEA_segments.csv* was generated previously from data recorded by the MEA software applied to 19 videos of therapy sessions. Three regions of interest (roi) were considered: head, torso and legs. The videos were split in 1 minute segments, one for each roi. In each segment the maximum and minimum Spearman cross-correlations were recorded (with maximal lags of 15 seconds in each direction). Only the segments where the movement of both subjects was above the median is kept.

Upload the data and select the relevant variables for the present analysis.

The database

The database *MEA_segments.csv* was generated previously from data recorded by the MEA software applied to 19 videos of therapy sessions. Three regions of interest (roi) were considered: head, torso and legs. The videos were split in 1 minute segments, one for each roi. In each segment the maximum and minimum Spearman cross-correlations were recorded (with maximal lags of 15 seconds in each direction). Only the segments where the movement of both subjects was above the median is kept.

Upload the data and select the relevant variables for the present analysis.

```r
library(dplyr)
```

```
##
## Adjuntando el paquete: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
```

```
##     intersect, setdiff, setequal, union
```

```
library(rstatix)
```

```
##
## Adjuntando el paquete: 'rstatix'

## The following object is masked from 'package:stats':
##
##     filter
```

```
library(ggplot2)
library(ggpubr)
library(knitr)
library(patchwork)
library(tidyr)

MEA_segments<-read.csv("MEA_segments.csv")%>%
  select(video,roi=zone,spearman_max,lag_spearman_max,spearman_min,lag_spearman_min,sex,type)
```

The data has a total of 445 observations.

The variables *spearman_max* and *spearman_min* are the extreme values attained by the spearman cross correlations for each observation. The variables __lag__spearman__max and *lag_spearman_min* give the time gap between the movement of each subject for which the extreme values were attained. A positive sign of the lag corresponds to the therapist movement preceding the patient.

PART 1: Influence of diagnosis on the strength of the correlation

For this part, the data was filtered to keep only the observations where the movement is led by the therapist. Also, lags with less to 1 second are discarded. This is done in order to avoid external noise that may give spurious simultaneous movement of the subjects. The data is split in two databases, in order to analyse the positive and negative correlations separately.

```
x_pos<-which(MEA_segments$spearman_max>0&MEA_segments$lag_spearman_max>1)
x_neg<-which(MEA_segments$spearman_min<0&MEA_segments$lag_spearman_min>1)

MEA_poscor<-MEA_segments[x_pos, ]%>%
  select(-spearman_min,-lag_spearman_min)
MEA_negcor<-MEA_segments[x_neg, ]%>%
  select(-spearman_max,-lag_spearman_max)
MEA_segments<-MEA_segments[union(x_pos,x_neg), ]
```

This leaves a total of 360 remaining observations, corresponding to 1 minute segments with a fixed roi. Of those, some have only positive correlations, some have negative correlations and some have both. The numbers of each are given below.

```
print(c("Positive correlations",length(x_pos)))
```

```
## [1] "Positive correlations" "195"
```

```
print(c("Negative correlations",length(x_neg)))
```

```
## [1] "Negative correlations" "216"
```

```
print(c("Both types", length(intersect(x_pos,x_neg))))
```

```
## [1] "Both types" "51"
```

They are distributed into the different 19 videos as follows:

```r
print("Segments with positive correlations per video")
```

```
## [1] "Segments with positive correlations per video"
```

```r
MEA_poscor <- group_by(MEA_poscor,video)
segments_pos_videos <- reframe(MEA_poscor,segments=n())
kable(segments_pos_videos)
```

video	segments
Vid0	14
Vid12	11
Vid13	12
Vid14	5
Vid15	9
Vid16	13
Vid17	6
Vid18	14
Vid19	12
Vid20	12
Vid21	8
Vid22	3
Vid23	3
Vid24	13
Vid25	14
Vid26	17
Vid27	7
Vid28	15
Vid29	7

```r
MEA_poscor<-ungroup(MEA_poscor)

print("Segments with negative correlations per video")
```

```
## [1] "Segments with negative correlations per video"
```

```r
MEA_negcor <- group_by(MEA_negcor,video)
segments_neg_videos <- reframe(MEA_negcor,segments=n())
kable(segments_neg_videos)
```

video	segments
Vid0	9
Vid12	3
Vid13	15
Vid14	9
Vid15	12
Vid16	7
Vid17	9
Vid18	10
Vid19	10
Vid20	9
Vid21	12
Vid22	13

video	segments
Vid23	10
Vid24	21
Vid25	8
Vid26	22
Vid27	12
Vid28	14
Vid29	11

```
MEA_negcor<-ungroup(MEA_negcor)

print("Complete database of positive correlations where the therapist leads")
```

```
## [1] "Complete database of positive correlations where the therapist leads"
```

```
kable(MEA_poscor)
```

video	roi	spearman_max	lag_spearman_max	sex	type
Vid0	head	0.0932581	9.10	Male	Treatment
Vid0	head	0.1418647	9.53	Male	Treatment
Vid0	head	0.1739299	13.07	Male	Treatment
Vid0	head	0.1753803	14.23	Male	Treatment
Vid0	head	0.1140846	10.30	Male	Treatment
Vid0	head	0.2511707	14.40	Male	Treatment
Vid0	torso	0.3293645	11.70	Male	Treatment
Vid0	torso	0.2884183	10.80	Male	Treatment
Vid0	torso	0.3456880	12.63	Male	Treatment
Vid0	torso	0.3843562	1.40	Male	Treatment
Vid0	legs	0.0960274	8.37	Male	Treatment
Vid0	legs	0.1790043	14.20	Male	Treatment
Vid0	legs	0.4123885	10.60	Male	Treatment
Vid0	legs	0.2542090	10.67	Male	Treatment
Vid12	head	0.2389910	6.87	Female	Treatment
Vid12	head	0.3789741	5.07	Female	Treatment
Vid12	head	0.5705451	10.33	Female	Treatment
Vid12	head	0.4084357	7.53	Female	Treatment
Vid12	head	0.4557194	11.53	Female	Treatment
Vid12	torso	0.4322892	5.33	Female	Treatment
Vid12	torso	0.4847488	5.60	Female	Treatment
Vid12	torso	0.2958076	14.27	Female	Treatment
Vid12	torso	0.4009657	12.93	Female	Treatment
Vid12	torso	0.3348079	4.20	Female	Treatment
Vid12	legs	0.1215066	7.37	Female	Treatment
Vid13	head	0.0092739	12.97	Female	Treatment
Vid13	head	0.2901365	14.33	Female	Treatment
Vid13	head	0.3935720	5.43	Female	Treatment
Vid13	head	0.3221192	1.37	Female	Treatment
Vid13	head	0.0326193	11.77	Female	Treatment
Vid13	head	0.1633125	5.90	Female	Treatment
Vid13	head	0.4264621	5.97	Female	Treatment
Vid13	torso	0.1896965	15.00	Female	Treatment
Vid13	torso	0.5053567	6.73	Female	Treatment

video	roi	spearman_max	lag_spearman_max	sex	type
Vid13	legs	0.1104169	10.47	Female	Treatment
Vid13	legs	0.2805974	12.00	Female	Treatment
Vid13	legs	0.2562024	11.13	Female	Treatment
Vid14	head	0.2799043	8.43	Female	Control
Vid14	torso	0.0403696	15.00	Female	Control
Vid14	legs	0.2043685	1.17	Female	Control
Vid14	legs	0.1640668	3.13	Female	Control
Vid14	legs	0.4364737	7.37	Female	Control
Vid15	head	0.3013106	11.57	Female	Control
Vid15	head	0.2164334	15.00	Female	Control
Vid15	head	0.2593998	2.50	Female	Control
Vid15	head	0.4954877	13.17	Female	Control
Vid15	head	0.3646107	8.30	Female	Control
Vid15	head	0.2366878	1.07	Female	Control
Vid15	torso	0.1883006	8.87	Female	Control
Vid15	torso	0.3170130	11.90	Female	Control
Vid15	legs	0.2550217	6.73	Female	Control
Vid16	head	0.1951704	9.50	Female	Treatment
Vid16	head	0.4147981	3.83	Female	Treatment
Vid16	head	0.3016753	9.37	Female	Treatment
Vid16	head	0.2324562	5.23	Female	Treatment
Vid16	head	0.2360189	9.07	Female	Treatment
Vid16	head	0.3090694	5.30	Female	Treatment
Vid16	torso	0.4017357	11.60	Female	Treatment
Vid16	torso	0.2990032	15.00	Female	Treatment
Vid16	legs	0.2669866	1.60	Female	Treatment
Vid16	legs	0.3044420	12.53	Female	Treatment
Vid16	legs	0.3838545	10.57	Female	Treatment
Vid16	legs	0.0973398	6.87	Female	Treatment
Vid16	legs	0.1859950	8.87	Female	Treatment
Vid17	head	0.5718205	11.37	Female	Control
Vid17	head	0.2498510	1.53	Female	Control
Vid17	torso	0.2965707	12.10	Female	Control
Vid17	torso	0.6106909	11.43	Female	Control
Vid17	torso	0.3918779	5.20	Female	Control
Vid17	legs	0.3637494	9.00	Female	Control
Vid18	head	0.0593795	10.43	Male	Control
Vid18	head	0.1853568	3.80	Male	Control
Vid18	head	0.2240068	11.87	Male	Control
Vid18	head	0.3134758	1.23	Male	Control
Vid18	torso	0.2115202	12.97	Male	Control
Vid18	torso	0.3936818	8.80	Male	Control
Vid18	torso	0.2360374	1.03	Male	Control
Vid18	legs	0.3246993	1.03	Male	Control
Vid18	legs	0.1450262	5.20	Male	Control
Vid18	legs	0.1642770	1.47	Male	Control
Vid18	legs	0.3135907	15.00	Male	Control
Vid18	legs	0.1488250	14.57	Male	Control
Vid18	legs	0.2589074	12.30	Male	Control
Vid18	legs	0.3053875	10.23	Male	Control
Vid19	head	0.3869406	6.53	Male	Treatment
Vid19	head	0.2138684	8.37	Male	Treatment

video	roi	spearman__max	lag__spearman__max	sex	type
Vid19	head	0.3010953	2.57	Male	Treatment
Vid19	head	0.1420079	15.00	Male	Treatment
Vid19	torso	0.4709510	1.43	Male	Treatment
Vid19	torso	0.5051660	8.50	Male	Treatment
Vid19	torso	0.1775441	8.20	Male	Treatment
Vid19	torso	0.3245651	12.17	Male	Treatment
Vid19	torso	0.2170910	10.63	Male	Treatment
Vid19	legs	0.2959564	1.33	Male	Treatment
Vid19	legs	0.2331549	11.03	Male	Treatment
Vid19	legs	0.3925859	4.13	Male	Treatment
Vid20	head	0.1904344	10.83	Female	Treatment
Vid20	head	0.5795734	13.83	Female	Treatment
Vid20	head	0.2598844	4.63	Female	Treatment
Vid20	torso	0.2996618	12.90	Female	Treatment
Vid20	torso	0.3107510	4.73	Female	Treatment
Vid20	torso	0.3300259	3.47	Female	Treatment
Vid20	legs	0.4168524	5.93	Female	Treatment
Vid20	legs	0.4527672	15.00	Female	Treatment
Vid20	legs	0.3293213	9.70	Female	Treatment
Vid20	legs	0.2496398	10.43	Female	Treatment
Vid20	legs	0.0460572	9.80	Female	Treatment
Vid20	legs	0.4738816	2.30	Female	Treatment
Vid21	head	0.2646839	6.20	Female	Treatment
Vid21	head	0.4641806	7.73	Female	Treatment
Vid21	head	0.1556868	13.80	Female	Treatment
Vid21	head	0.5160339	10.13	Female	Treatment
Vid21	head	0.3622348	7.50	Female	Treatment
Vid21	torso	0.1649575	15.00	Female	Treatment
Vid21	legs	0.2054637	1.20	Female	Treatment
Vid21	legs	0.4255196	11.03	Female	Treatment
Vid22	torso	0.2737741	14.43	Female	Control
Vid22	legs	0.1797252	15.00	Female	Control
Vid22	legs	0.2367898	2.00	Female	Control
Vid23	torso	0.2616188	13.87	Female	Treatment
Vid23	legs	0.4092809	2.80	Female	Treatment
Vid23	legs	0.3991437	2.33	Female	Treatment
Vid24	head	0.2970765	14.97	Female	Control
Vid24	head	0.1529493	15.00	Female	Control
Vid24	head	0.5295959	12.00	Female	Control
Vid24	head	0.3944789	5.00	Female	Control
Vid24	torso	0.1770872	14.57	Female	Control
Vid24	torso	0.1186835	13.23	Female	Control
Vid24	torso	0.2127131	11.87	Female	Control
Vid24	legs	0.1997720	9.93	Female	Control
Vid24	legs	0.2314877	1.20	Female	Control
Vid24	legs	0.3259892	4.13	Female	Control
Vid24	legs	0.4039745	2.33	Female	Control
Vid24	legs	0.4683443	15.00	Female	Control
Vid24	legs	0.4272356	1.80	Female	Control
Vid25	head	0.2212264	15.00	Male	Control
Vid25	head	0.2642308	11.40	Male	Control
Vid25	head	0.0559476	15.00	Male	Control

video	roi	spearman__max	lag__spearman__max	sex	type
Vid25	head	0.1837116	15.00	Male	Control
Vid25	head	0.1130602	15.00	Male	Control
Vid25	torso	0.2079676	3.27	Male	Control
Vid25	torso	0.0605300	14.87	Male	Control
Vid25	torso	0.0958539	14.23	Male	Control
Vid25	legs	0.3054408	3.67	Male	Control
Vid25	legs	0.4926588	7.47	Male	Control
Vid25	legs	0.4754431	1.40	Male	Control
Vid25	legs	0.2074146	1.73	Male	Control
Vid25	legs	0.2665044	15.00	Male	Control
Vid25	legs	0.2882369	15.00	Male	Control
Vid26	head	0.4352934	8.57	Female	Treatment
Vid26	head	0.3637812	9.50	Female	Treatment
Vid26	head	0.2362560	14.63	Female	Treatment
Vid26	head	0.2814595	12.00	Female	Treatment
Vid26	head	0.2714993	5.67	Female	Treatment
Vid26	head	0.1326967	15.00	Female	Treatment
Vid26	torso	0.4605616	11.47	Female	Treatment
Vid26	torso	0.2319369	2.63	Female	Treatment
Vid26	legs	0.1790326	5.93	Female	Treatment
Vid26	legs	0.2733583	9.50	Female	Treatment
Vid26	legs	0.3510698	3.93	Female	Treatment
Vid26	legs	0.3657932	4.23	Female	Treatment
Vid26	legs	0.3309416	5.53	Female	Treatment
Vid26	legs	0.4571363	1.07	Female	Treatment
Vid26	legs	0.1967172	1.07	Female	Treatment
Vid26	legs	0.5635275	6.87	Female	Treatment
Vid26	legs	0.3750125	1.97	Female	Treatment
Vid27	head	0.2620423	7.00	Male	Treatment
Vid27	head	0.1398915	14.90	Male	Treatment
Vid27	torso	0.2714596	10.23	Male	Treatment
Vid27	legs	0.2121436	15.00	Male	Treatment
Vid27	legs	0.3113877	10.70	Male	Treatment
Vid27	legs	0.2728723	9.50	Male	Treatment
Vid27	legs	0.3143476	12.67	Male	Treatment
Vid28	head	0.2433795	14.80	Female	Treatment
Vid28	head	0.2064252	1.47	Female	Treatment
Vid28	head	0.2002257	15.00	Female	Treatment
Vid28	head	0.2719629	1.20	Female	Treatment
Vid28	head	0.2870553	9.17	Female	Treatment
Vid28	head	0.2860443	6.37	Female	Treatment
Vid28	head	0.5075627	5.80	Female	Treatment
Vid28	torso	0.3302227	11.80	Female	Treatment
Vid28	torso	0.4352305	13.57	Female	Treatment
Vid28	torso	0.1623488	13.30	Female	Treatment
Vid28	legs	0.2235145	15.00	Female	Treatment
Vid28	legs	0.2353918	7.57	Female	Treatment
Vid28	legs	0.2879363	5.90	Female	Treatment
Vid28	legs	0.2185400	9.47	Female	Treatment
Vid28	legs	0.3861260	1.60	Female	Treatment
Vid29	head	0.2491301	13.93	Female	Treatment
Vid29	head	0.1872974	15.00	Female	Treatment

video	roi	spearman_max	lag_spearman_max	sex	type
Vid29	torso	0.3088549	15.00	Female	Treatment
Vid29	torso	0.3102463	8.60	Female	Treatment
Vid29	legs	0.0454432	14.57	Female	Treatment
Vid29	legs	0.2292280	10.10	Female	Treatment
Vid29	legs	0.2281292	13.07	Female	Treatment

```r
print("Complete database of negative correlations where the therapist leads")
```

```
## [1] "Complete database of negative correlations where the therapist leads"
```

```r
kable(MEA_negcor)
```

video	roi	spearman_min	lag_spearman_min	sex	type
Vid0	head	-0.2510538	9.67	Male	Treatment
Vid0	head	-0.4834532	2.03	Male	Treatment
Vid0	head	-0.2362128	15.00	Male	Treatment
Vid0	head	-0.4894361	2.50	Male	Treatment
Vid0	torso	-0.3532904	12.23	Male	Treatment
Vid0	torso	-0.1255647	14.87	Male	Treatment
Vid0	torso	-0.5121500	1.47	Male	Treatment
Vid0	torso	-0.3038312	15.00	Male	Treatment
Vid0	legs	-0.2637888	13.27	Male	Treatment
Vid12	head	-0.0778809	15.00	Female	Treatment
Vid12	legs	-0.0128393	15.00	Female	Treatment
Vid12	legs	-0.1720222	8.17	Female	Treatment
Vid13	head	-0.2365106	2.70	Female	Treatment
Vid13	head	-0.2432960	15.00	Female	Treatment
Vid13	head	-0.4070916	6.60	Female	Treatment
Vid13	head	-0.2979437	10.77	Female	Treatment
Vid13	head	-0.1101470	14.37	Female	Treatment
Vid13	torso	-0.5391773	7.07	Female	Treatment
Vid13	torso	-0.3165275	13.90	Female	Treatment
Vid13	torso	-0.4238474	3.13	Female	Treatment
Vid13	torso	-0.2609763	6.97	Female	Treatment
Vid13	torso	-0.2254984	15.00	Female	Treatment
Vid13	torso	-0.3467287	15.00	Female	Treatment
Vid13	torso	-0.5349092	1.57	Female	Treatment
Vid13	torso	-0.1616789	15.00	Female	Treatment
Vid13	legs	-0.3089026	5.10	Female	Treatment
Vid13	legs	-0.2452251	15.00	Female	Treatment
Vid14	head	-0.2036692	5.93	Female	Control
Vid14	head	-0.3558005	9.23	Female	Control
Vid14	head	-0.0886599	15.00	Female	Control
Vid14	torso	-0.3216319	10.33	Female	Control
Vid14	torso	-0.6681625	1.23	Female	Control
Vid14	torso	-0.6451860	2.20	Female	Control
Vid14	torso	-0.3844945	9.67	Female	Control
Vid14	legs	-0.2393221	2.23	Female	Control
Vid14	legs	-0.3502958	6.40	Female	Control
Vid15	head	-0.2245214	3.50	Female	Control
Vid15	head	-0.2244296	7.63	Female	Control

video	roi	spearman_min	lag_spearman_min	sex	type
Vid15	head	-0.3251316	4.87	Female	Control
Vid15	head	-0.3095610	4.43	Female	Control
Vid15	head	-0.2545530	1.90	Female	Control
Vid15	head	-0.1045769	12.87	Female	Control
Vid15	torso	-0.5021249	3.43	Female	Control
Vid15	torso	-0.2876773	15.00	Female	Control
Vid15	legs	-0.4209090	15.00	Female	Control
Vid15	legs	-0.1610942	8.10	Female	Control
Vid15	legs	-0.5925171	2.17	Female	Control
Vid15	legs	-0.2801005	12.20	Female	Control
Vid16	head	-0.1543152	14.07	Female	Treatment
Vid16	head	-0.2705761	8.37	Female	Treatment
Vid16	torso	-0.2959293	3.27	Female	Treatment
Vid16	torso	-0.2944953	15.00	Female	Treatment
Vid16	legs	-0.2912475	10.60	Female	Treatment
Vid16	legs	-0.3419917	3.63	Female	Treatment
Vid16	legs	-0.4617461	4.00	Female	Treatment
Vid17	head	-0.2942240	12.07	Female	Control
Vid17	head	-0.5744461	3.27	Female	Control
Vid17	head	-0.2569905	1.43	Female	Control
Vid17	head	-0.1971874	4.80	Female	Control
Vid17	head	-0.2137783	15.00	Female	Control
Vid17	torso	-0.3745304	7.87	Female	Control
Vid17	torso	-0.5208819	3.20	Female	Control
Vid17	legs	-0.4033656	11.40	Female	Control
Vid17	legs	-0.4659171	3.63	Female	Control
Vid18	head	-0.2370786	5.57	Male	Control
Vid18	torso	-0.2763742	6.07	Male	Control
Vid18	torso	-0.4322449	5.17	Male	Control
Vid18	torso	-0.2571632	1.63	Male	Control
Vid18	torso	-0.2302350	12.83	Male	Control
Vid18	torso	-0.2186759	6.73	Male	Control
Vid18	legs	-0.1914383	15.00	Male	Control
Vid18	legs	-0.2616739	13.90	Male	Control
Vid18	legs	-0.3510255	4.93	Male	Control
Vid18	legs	-0.2639537	8.33	Male	Control
Vid19	head	-0.1908735	4.70	Male	Treatment
Vid19	head	-0.3915656	13.00	Male	Treatment
Vid19	head	-0.0992456	11.03	Male	Treatment
Vid19	head	-0.1272868	10.33	Male	Treatment
Vid19	torso	-0.3964552	11.73	Male	Treatment
Vid19	torso	-0.3627017	12.10	Male	Treatment
Vid19	torso	-0.2861578	3.93	Male	Treatment
Vid19	legs	-0.5081251	8.93	Male	Treatment
Vid19	legs	-0.3545276	14.70	Male	Treatment
Vid19	legs	-0.2018961	9.63	Male	Treatment
Vid20	head	-0.3023598	7.87	Female	Treatment
Vid20	head	-0.2513814	1.57	Female	Treatment
Vid20	head	-0.3838958	9.47	Female	Treatment
Vid20	torso	-0.2659066	4.87	Female	Treatment
Vid20	torso	-0.3994606	10.63	Female	Treatment
Vid20	legs	-0.1441970	2.17	Female	Treatment

video	roi	spearman_min	lag_spearman_min	sex	type
Vid20	legs	-0.3743465	12.07	Female	Treatment
Vid20	legs	-0.2833259	13.87	Female	Treatment
Vid20	legs	-0.3054469	15.00	Female	Treatment
Vid21	head	-0.2292545	3.27	Female	Treatment
Vid21	head	-0.0946742	4.90	Female	Treatment
Vid21	head	-0.2962030	3.30	Female	Treatment
Vid21	head	-0.1115884	9.00	Female	Treatment
Vid21	head	-0.1919576	9.67	Female	Treatment
Vid21	head	-0.2908725	11.63	Female	Treatment
Vid21	torso	-0.0344602	13.37	Female	Treatment
Vid21	torso	-0.4872609	1.53	Female	Treatment
Vid21	torso	-0.3846231	7.30	Female	Treatment
Vid21	torso	-0.2331779	15.00	Female	Treatment
Vid21	torso	-0.2109840	15.00	Female	Treatment
Vid21	legs	-0.0531587	8.90	Female	Treatment
Vid22	head	-0.3052169	14.27	Female	Control
Vid22	head	-0.4894697	9.03	Female	Control
Vid22	head	-0.4077560	3.27	Female	Control
Vid22	head	-0.1452458	12.73	Female	Control
Vid22	head	-0.3587086	6.67	Female	Control
Vid22	torso	-0.5376649	4.43	Female	Control
Vid22	torso	-0.5831122	3.67	Female	Control
Vid22	torso	-0.5661414	8.67	Female	Control
Vid22	torso	-0.4476040	15.00	Female	Control
Vid22	legs	-0.2524925	1.50	Female	Control
Vid22	legs	-0.3362006	6.53	Female	Control
Vid22	legs	-0.3768838	8.33	Female	Control
Vid22	legs	-0.4496626	12.57	Female	Control
Vid23	head	-0.2304007	10.87	Female	Treatment
Vid23	torso	-0.2658581	11.67	Female	Treatment
Vid23	torso	-0.3815664	11.97	Female	Treatment
Vid23	torso	-0.2488419	4.77	Female	Treatment
Vid23	torso	-0.6882584	2.50	Female	Treatment
Vid23	torso	-0.3224015	14.03	Female	Treatment
Vid23	legs	-0.3434872	6.27	Female	Treatment
Vid23	legs	-0.2639307	12.63	Female	Treatment
Vid23	legs	-0.1973385	11.53	Female	Treatment
Vid23	legs	-0.1105949	8.77	Female	Treatment
Vid24	head	-0.3758879	1.87	Female	Control
Vid24	head	-0.2608245	2.67	Female	Control
Vid24	head	-0.3143877	1.13	Female	Control
Vid24	head	-0.5598496	4.10	Female	Control
Vid24	head	-0.5486247	4.13	Female	Control
Vid24	head	-0.2410617	2.80	Female	Control
Vid24	head	-0.4684841	7.00	Female	Control
Vid24	torso	-0.5202572	6.03	Female	Control
Vid24	torso	-0.2451455	8.77	Female	Control
Vid24	torso	-0.4440730	6.17	Female	Control
Vid24	torso	-0.3970452	3.23	Female	Control
Vid24	torso	-0.2908285	1.37	Female	Control
Vid24	torso	-0.6954566	4.10	Female	Control
Vid24	torso	-0.3221397	5.93	Female	Control

video	roi	spearman_min	lag_spearman_min	sex	type
Vid24	legs	-0.4749411	6.37	Female	Control
Vid24	legs	-0.2839440	9.70	Female	Control
Vid24	legs	-0.4318160	3.40	Female	Control
Vid24	legs	-0.3614938	12.63	Female	Control
Vid24	legs	-0.4025627	11.23	Female	Control
Vid24	legs	-0.2300123	8.73	Female	Control
Vid24	legs	-0.3521329	11.97	Female	Control
Vid25	head	-0.1459933	14.80	Male	Control
Vid25	torso	-0.3894376	5.80	Male	Control
Vid25	torso	-0.3177209	1.27	Male	Control
Vid25	legs	-0.2306857	5.93	Male	Control
Vid25	legs	-0.2792696	6.20	Male	Control
Vid25	legs	-0.1493671	15.00	Male	Control
Vid25	legs	-0.1492407	8.87	Male	Control
Vid25	legs	-0.1057316	15.00	Male	Control
Vid26	head	-0.2718387	3.10	Female	Treatment
Vid26	head	-0.2611763	8.37	Female	Treatment
Vid26	head	-0.3478540	13.93	Female	Treatment
Vid26	head	-0.3543979	3.23	Female	Treatment
Vid26	head	-0.3532213	6.37	Female	Treatment
Vid26	torso	-0.4270781	4.17	Female	Treatment
Vid26	torso	-0.2393715	9.83	Female	Treatment
Vid26	torso	-0.3999950	7.57	Female	Treatment
Vid26	torso	-0.4645893	13.90	Female	Treatment
Vid26	torso	-0.4622864	4.30	Female	Treatment
Vid26	torso	-0.2973012	8.50	Female	Treatment
Vid26	legs	-0.3189350	6.80	Female	Treatment
Vid26	legs	-0.1395029	3.50	Female	Treatment
Vid26	legs	-0.3620815	6.13	Female	Treatment
Vid26	legs	-0.1858194	6.67	Female	Treatment
Vid26	legs	-0.3884098	9.57	Female	Treatment
Vid26	legs	-0.1588711	2.57	Female	Treatment
Vid26	legs	-0.1633838	14.43	Female	Treatment
Vid26	legs	-0.3293803	11.93	Female	Treatment
Vid26	legs	-0.5413812	5.33	Female	Treatment
Vid26	legs	-0.2796247	9.57	Female	Treatment
Vid26	legs	-0.1702061	7.70	Female	Treatment
Vid27	head	-0.2713167	9.77	Male	Treatment
Vid27	head	-0.2424107	12.87	Male	Treatment
Vid27	torso	-0.3778011	8.43	Male	Treatment
Vid27	torso	-0.2461194	5.57	Male	Treatment
Vid27	torso	-0.3050708	11.60	Male	Treatment
Vid27	torso	-0.3085388	6.30	Male	Treatment
Vid27	torso	-0.2865906	6.93	Male	Treatment
Vid27	legs	-0.3473131	15.00	Male	Treatment
Vid27	legs	-0.3347834	7.97	Male	Treatment
Vid27	legs	-0.3422575	1.83	Male	Treatment
Vid27	legs	-0.3876540	1.90	Male	Treatment
Vid27	legs	-0.3066379	6.53	Male	Treatment
Vid28	head	-0.3645014	4.30	Female	Treatment
Vid28	head	-0.4542677	2.83	Female	Treatment
Vid28	head	-0.2059703	3.47	Female	Treatment

video	roi	spearman_min	lag_spearman_min	sex	type
Vid28	head	-0.2472913	8.63	Female	Treatment
Vid28	head	-0.2121306	13.77	Female	Treatment
Vid28	torso	-0.0304252	13.30	Female	Treatment
Vid28	torso	-0.3485951	10.10	Female	Treatment
Vid28	torso	-0.2636383	14.37	Female	Treatment
Vid28	torso	-0.3750550	9.30	Female	Treatment
Vid28	torso	-0.4428723	6.37	Female	Treatment
Vid28	torso	-0.1601889	9.90	Female	Treatment
Vid28	legs	-0.0738424	13.50	Female	Treatment
Vid28	legs	-0.3130658	9.83	Female	Treatment
Vid28	legs	-0.2486158	4.43	Female	Treatment
Vid29	head	-0.3902335	2.50	Female	Treatment
Vid29	head	-0.3164998	2.73	Female	Treatment
Vid29	head	-0.5647934	6.67	Female	Treatment
Vid29	torso	-0.4324098	2.43	Female	Treatment
Vid29	torso	-0.2554002	8.03	Female	Treatment
Vid29	torso	-0.5022196	7.30	Female	Treatment
Vid29	torso	-0.2844432	10.23	Female	Treatment
Vid29	torso	-0.3941189	2.97	Female	Treatment
Vid29	torso	-0.5214886	4.23	Female	Treatment
Vid29	legs	-0.2703380	6.30	Female	Treatment
Vid29	legs	-0.3027828	14.30	Female	Treatment

```
MEA_poscor<-group_by(MEA_poscor,video,sex,type)
Mean_pcorr_by_video<-reframe(MEA_poscor,mean_pc=mean(spearman_max),sd_pc=sd(spearman_max),segments_pos=
MEA_poscor<-ungroup(MEA_poscor)

kable(Mean_pcorr_by_video)
```

video	sex	type	mean_pc	sd_pc	segments_pos
Vid0	Male	Treatment	0.2313675	0.1078989	14
Vid12	Female	Treatment	0.3747992	0.1235039	11
Vid13	Female	Treatment	0.2483138	0.1539036	12
Vid14	Female	Control	0.2250366	0.1465968	5
Vid15	Female	Control	0.2926961	0.0931001	9
Vid16	Female	Treatment	0.2791189	0.0911962	13
Vid17	Female	Control	0.4140934	0.1465221	6
Vid18	Male	Control	0.2345837	0.0901229	14
Vid19	Male	Treatment	0.3050772	0.1153956	12
Vid20	Female	Treatment	0.3282375	0.1408849	12
Vid21	Female	Treatment	0.3198451	0.1410931	8
Vid22	Female	Control	0.2300964	0.0473804	3
Vid23	Female	Treatment	0.3566812	0.0824823	3
Vid24	Female	Control	0.3030298	0.1321332	13
Vid25	Male	Control	0.2313019	0.1343632	14
Vid26	Female	Treatment	0.3238867	0.1145548	17
Vid27	Male	Treatment	0.2548778	0.0611394	7
Vid28	Female	Treatment	0.2854644	0.0946234	15
Vid29	Female	Treatment	0.2226185	0.0899360	7

Mean extremal negative correlations, for each video:

```
MEA_negcor<-group_by(MEA_negcor,video,sex,type)
Mean_ncorr_by_video<-reframe(MEA_negcor,mean_nc=mean(spearman_min),sd_nc=sd(spearman_min),segments_neg=n
MEA_negcor<-ungroup(MEA_negcor)

kable(Mean_ncorr_by_video)
```

video	sex	type	mean_nc	sd_nc	segments_neg
Vid0	Male	Treatment	-0.3354201	0.1343110	9
Vid12	Female	Treatment	-0.0875808	0.0800336	3
Vid13	Female	Treatment	-0.3105640	0.1232181	15
Vid14	Female	Control	-0.3619136	0.1907466	9
Vid15	Female	Control	-0.3072664	0.1390341	12
Vid16	Female	Treatment	-0.3014716	0.0913995	7
Vid17	Female	Control	-0.3668135	0.1358255	9
Vid18	Male	Control	-0.2719863	0.0704206	10
Vid19	Male	Treatment	-0.2918835	0.1329606	10
Vid20	Female	Treatment	-0.3011467	0.0794988	9
Vid21	Female	Treatment	-0.2181846	0.1348963	12
Vid22	Female	Control	-0.4043199	0.1273418	13
Vid23	Female	Treatment	-0.3052678	0.1548672	10
Vid24	Female	Control	-0.3914747	0.1228391	21
Vid25	Male	Control	-0.2209308	0.1002916	8
Vid26	Female	Treatment	-0.3143957	0.1101680	22
Vid27	Male	Treatment	-0.3130412	0.0469955	12
Vid28	Female	Treatment	-0.2671757	0.1265279	14
Vid29	Female	Treatment	-0.3849753	0.1088855	11

And per group and type cohorots, including number of segments.

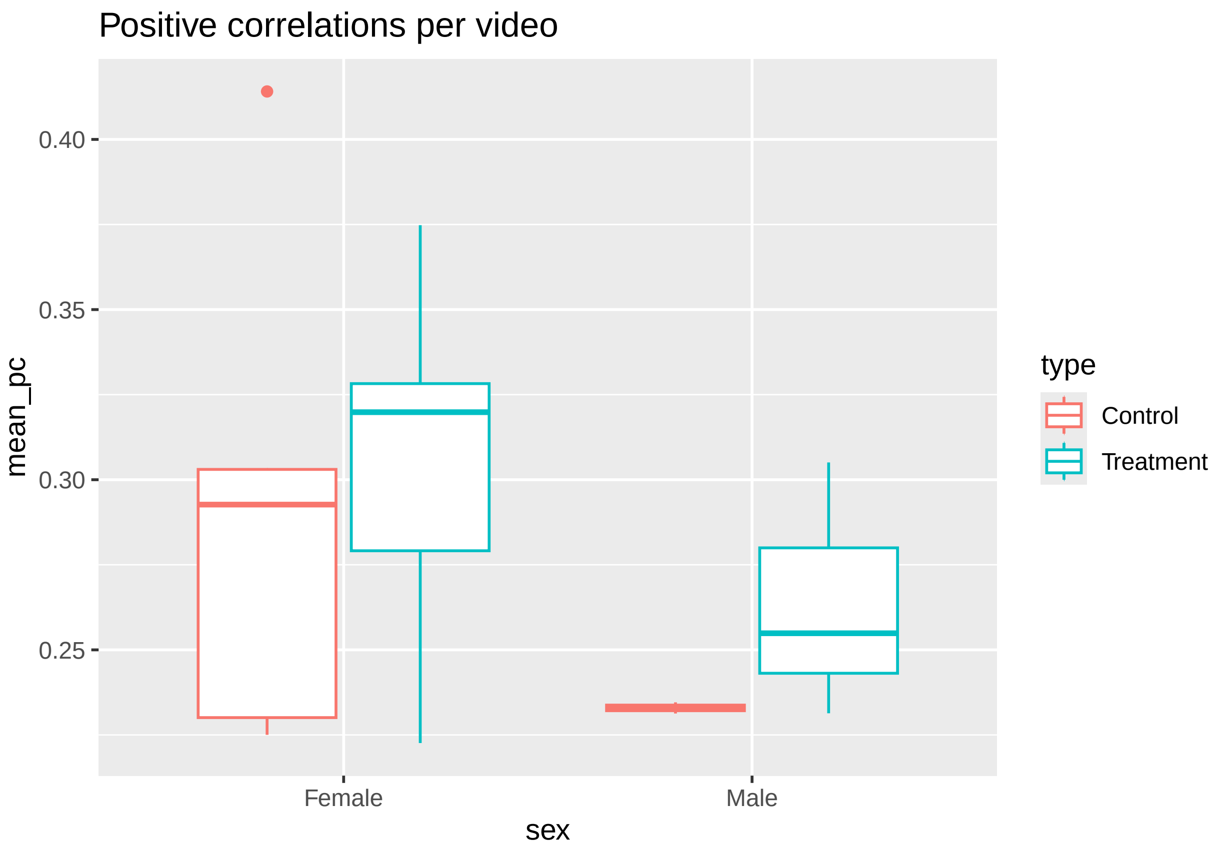
```
MEA_poscor<-group_by(MEA_poscor,sex,type)
Mean_pcorr<-reframe(MEA_poscor,mean_pc=mean(spearman_max),sd_pc=sd(spearman_max),segments_pos=n())
MEA_poscor<-ungroup(MEA_poscor)

MEA_negcor<-group_by(MEA_negcor,sex,type)
Mean_ncorr<-reframe(MEA_negcor,mean_nc=mean(spearman_min),sd_nc=sd(spearman_min),segments_neg=n())
MEA_negcor<-ungroup(MEA_negcor)

MEA_mean_cors<-merge(Mean_pcorr,Mean_ncorr,by=c("sex","type"))

kable(MEA_mean_cors)
```

sex	type	mean_pc	sd_pc	segments_pos	mean_nc	sd_nc	segments_neg
Female	Control	0.3020468	0.1307076	36	-0.3706698	0.1391899	64
Female	Treatment	0.3025013	0.1220030	98	-0.2942195	0.1266821	103
Male	Control	0.2329428	0.1122758	28	-0.2492950	0.0863050	18
Male	Treatment	0.2631581	0.1054706	33	-0.3127132	0.1059398	31

Visualization:

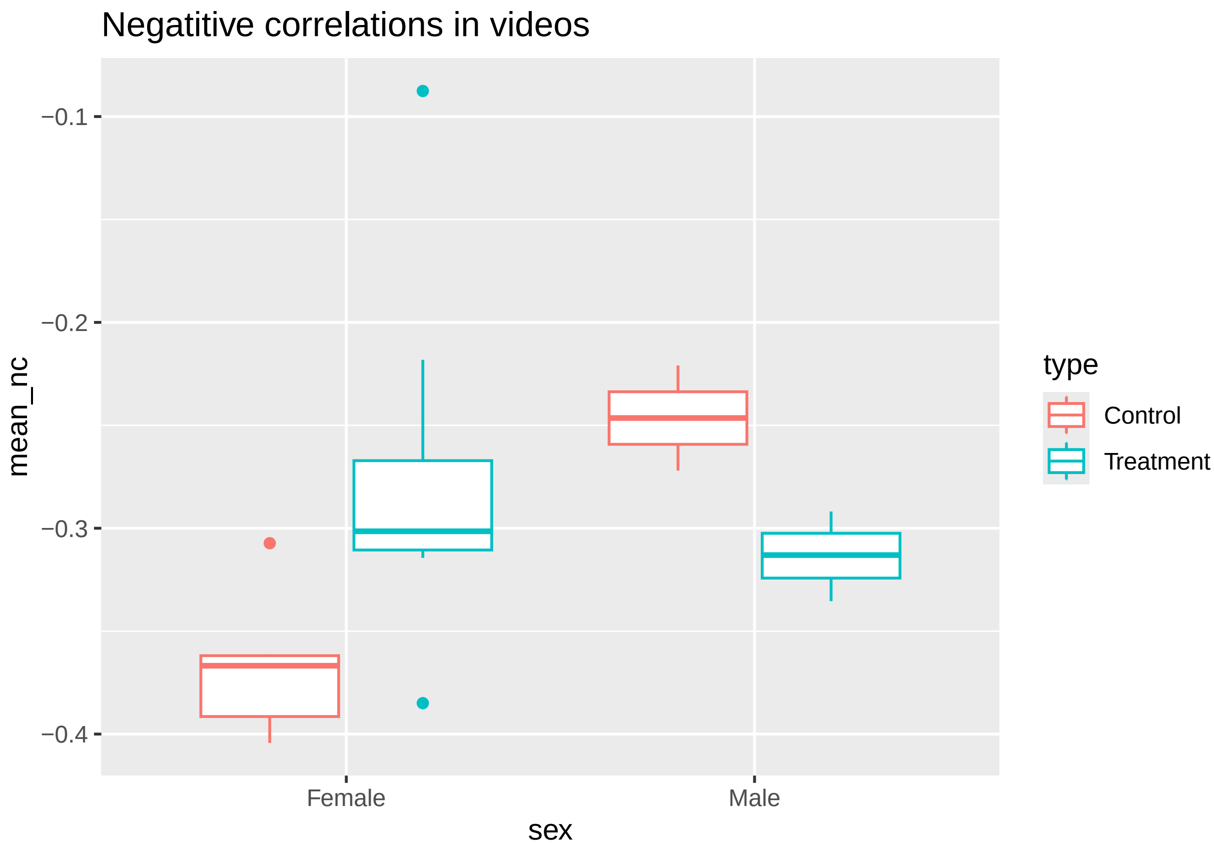
Positive correlations

```
ggplot(Mean_pcorr_by_video,aes(x=sex,y=mean_pc,color=type))+geom_boxplot()+ggtitle("Positive correlatio
```



Negative correlations

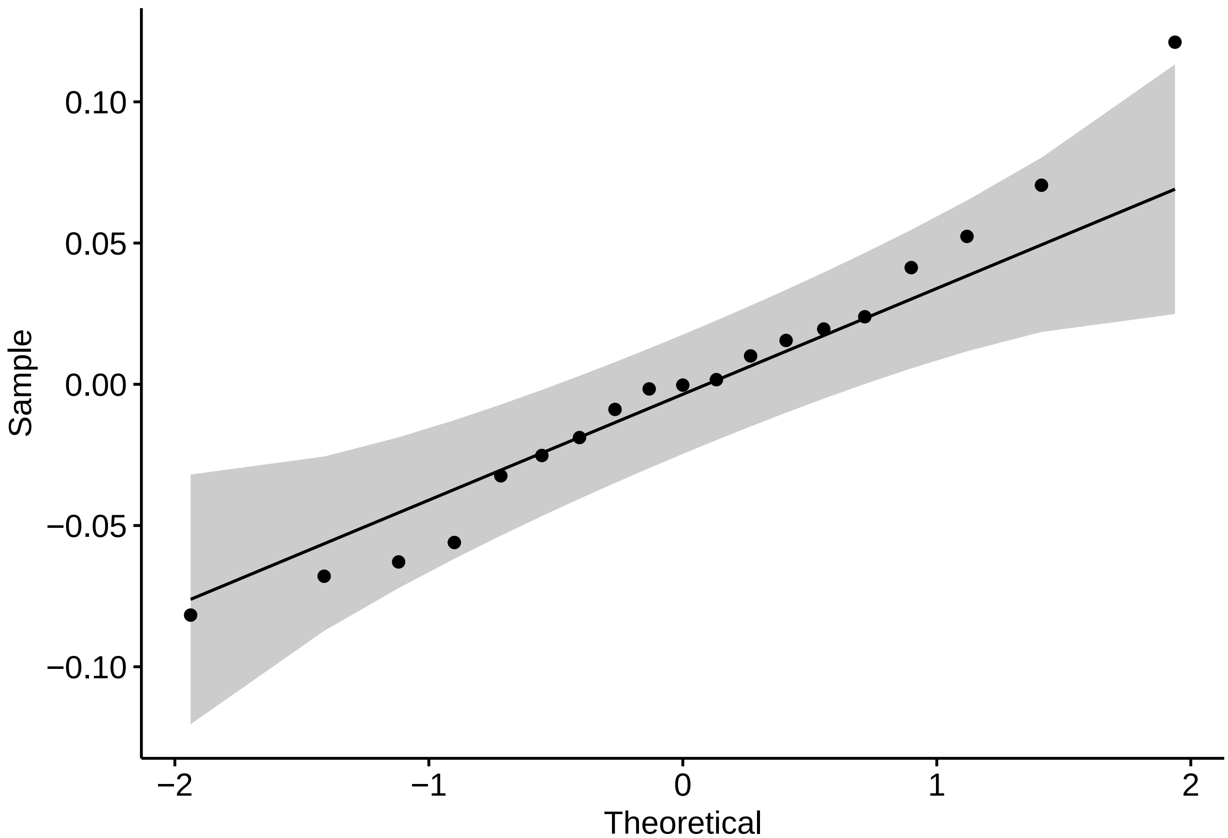
```
ggplot(Mean_ncorr_by_video,aes(x=sex,y=mean_nc,color=type))+geom_boxplot()+ggtitle("Negatitive correlat
```



Statistical analysis for PART 1.

Test for normality and homogeneity (positive correlations).

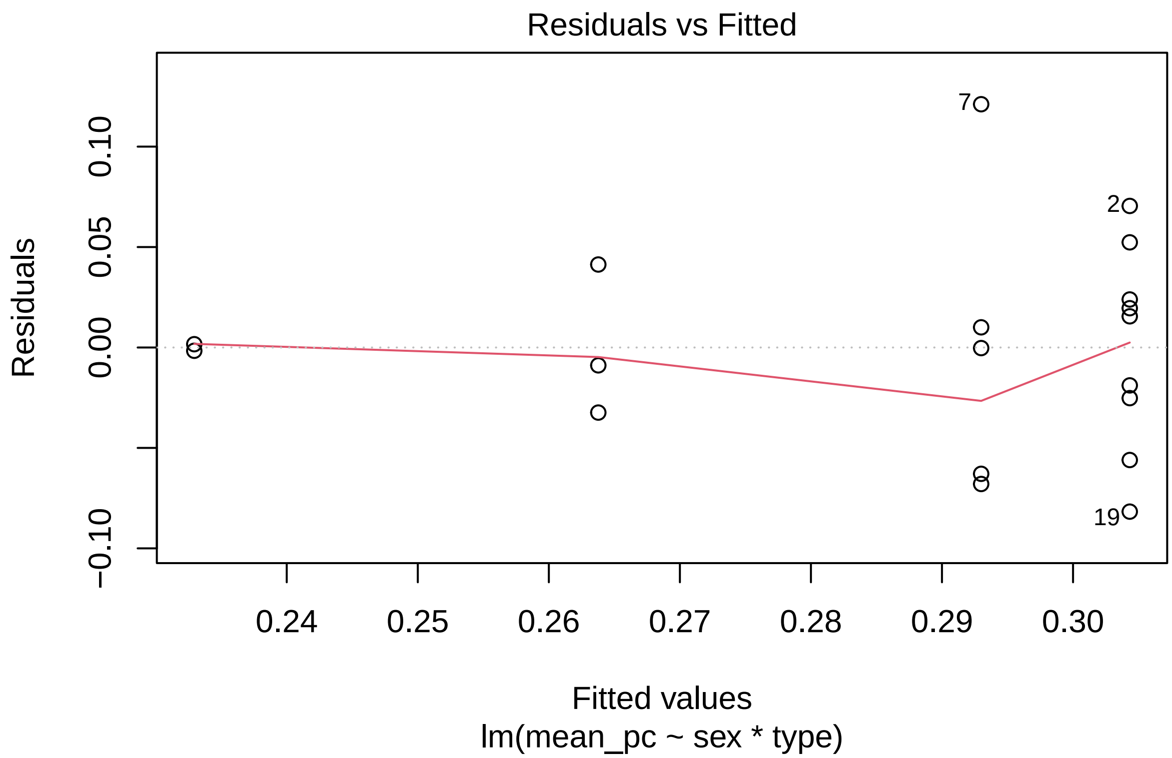
```
model  <- lm(mean_pc ~ sex*type, data = Mean_pcorr_by_video)
ggqqplot(residuals(model))
```




```r
shapiro_test(residuals(model))
```

```
## # A tibble: 1 x 3
##   variable         statistic p.value
##   <chr>                <dbl>   <dbl>
## 1 residuals(model)     0.972   0.812
```

```r
plot(model,1)
```

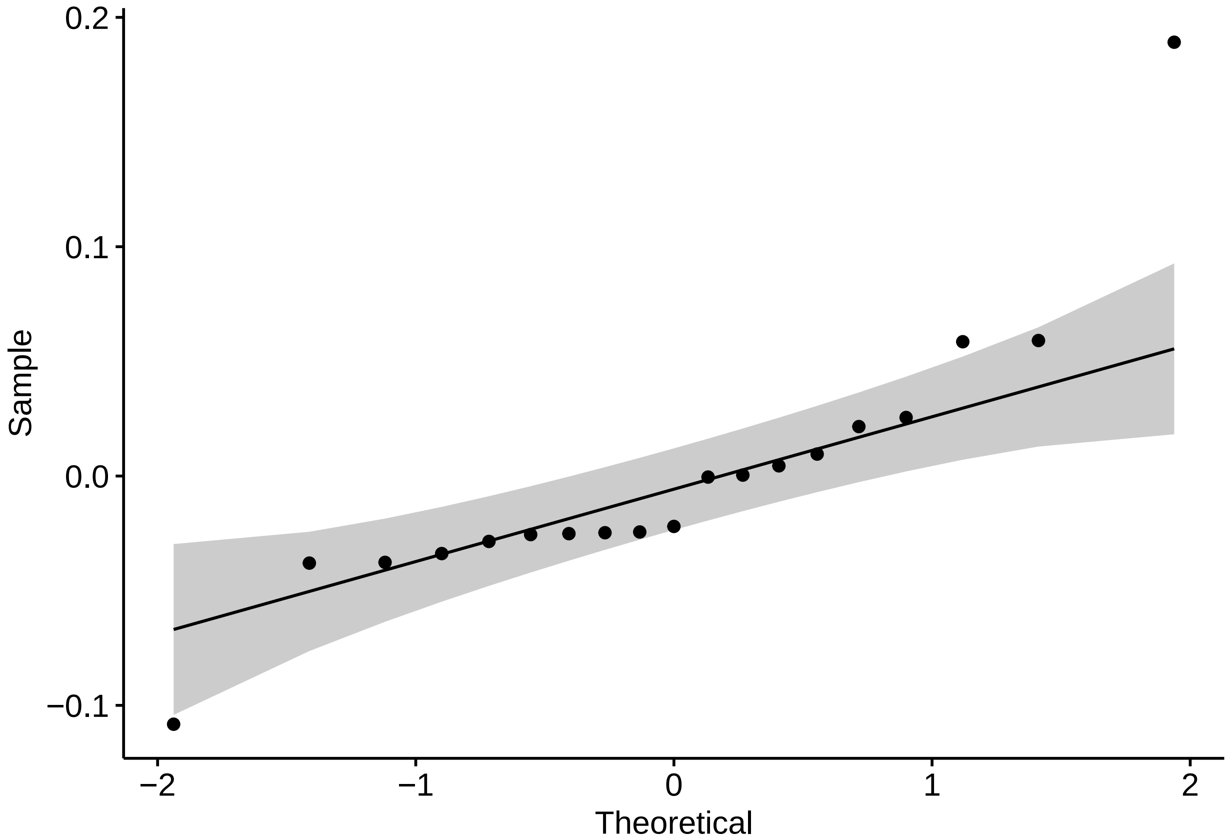



```
Mean_pcorr_by_video %>% levene_test(mean_pc ~ sex*type)
```

```
## # A tibble: 1 x 4
##     df1   df2 statistic     p
##   <int> <int>     <dbl> <dbl>
## 1     3    15      1.08 0.388
```

Test for normality and homogeneity (negative correlations).

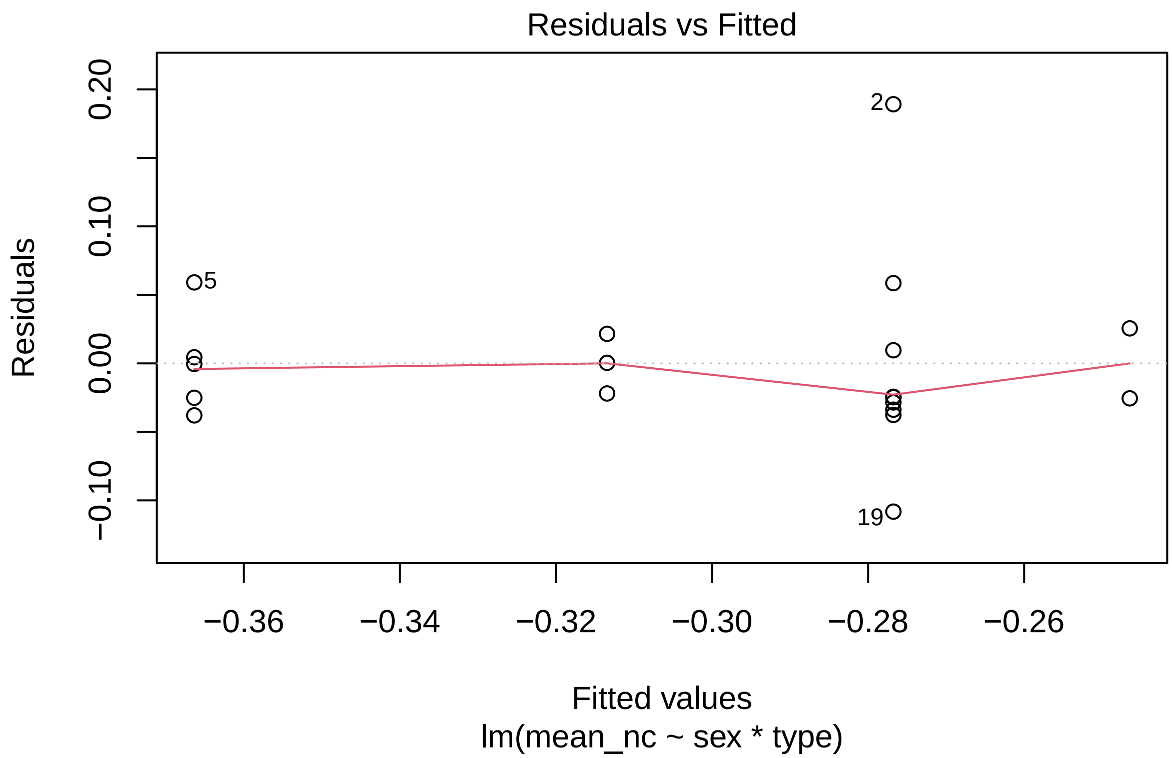
```
model  <- lm(mean_nc ~ sex*type, data = Mean_ncorr_by_video)
ggqqplot(residuals(model))
```




```r
shapiro_test(residuals(model))
```

```
## # A tibble: 1 x 3
##   variable         statistic p.value
##   <chr>                <dbl>   <dbl>
## 1 residuals(model)     0.826 0.00278
```

```r
plot(model,1)
```




```
Mean_ncorr_by_video %>% levene_test(mean_nc ~ sex*type)
```

```
## # A tibble: 1 x 4
##     df1   df2 statistic     p
##   <int> <int>     <dbl> <dbl>
## 1     3    15     0.436 0.731
```

Now, we look for outliers. For positive correlations:

```
temp<-Mean_pcorr_by_video%>%group_by(sex,type)%>%identify_outliers(mean_pc)
print(temp)
```

```
## # A tibble: 1 x 8
##   sex    type    video mean_pc sd_pc segments_pos is.outlier is.extreme
##   <chr>  <chr>   <chr>   <dbl> <dbl>        <int> <lgl>      <lgl>
## 1 Female Control Vid17   0.414 0.147            6 TRUE       FALSE
```

For negative correlations:

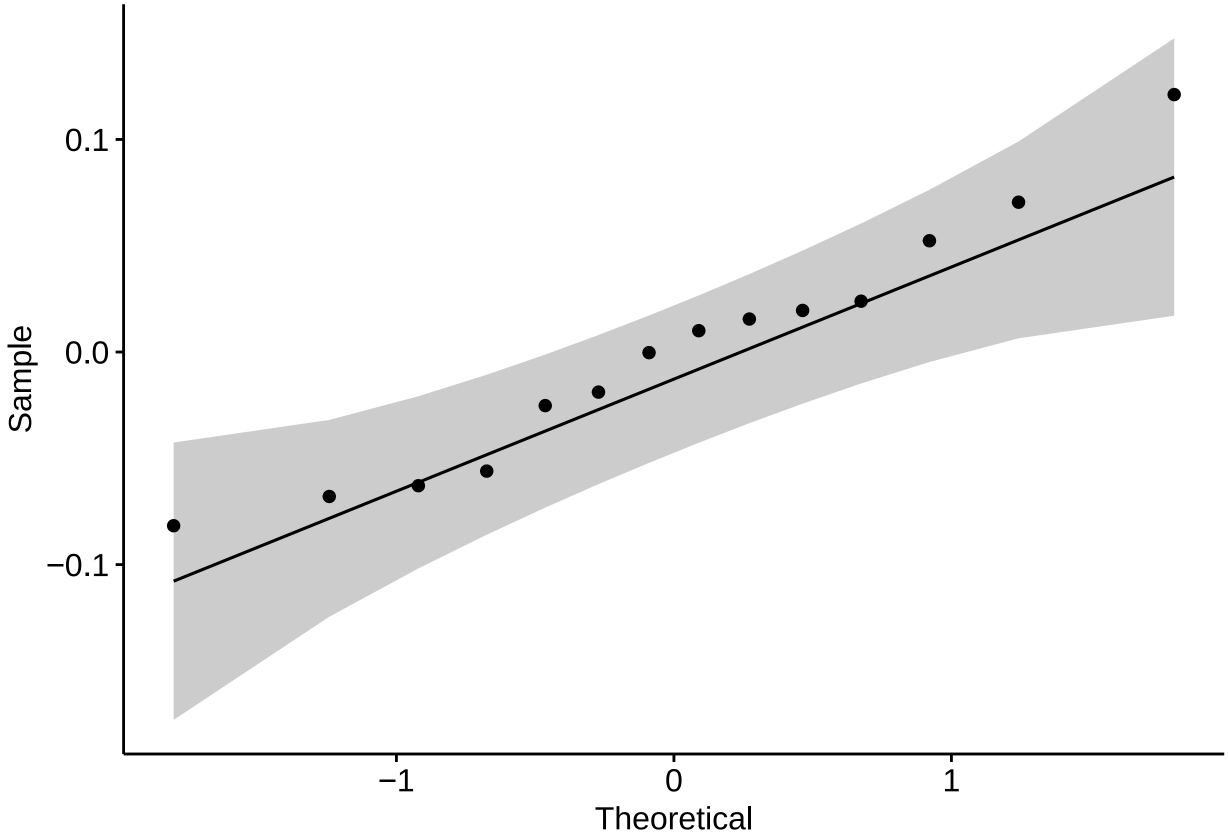
```
temp<-Mean_ncorr_by_video%>%group_by(sex,type)%>%identify_outliers(mean_nc)
print(temp)
```

```
## # A tibble: 3 x 8
##   sex    type      video mean_nc  sd_nc segments_neg is.outlier is.extreme
##   <chr>  <chr>     <chr>   <dbl>  <dbl>        <int> <lgl>      <lgl>
## 1 Female Control   Vid15 -0.307  0.139            12 TRUE       FALSE
## 2 Female Treatment Vid12 -0.0876 0.0800            3 TRUE       TRUE
## 3 Female Treatment Vid29 -0.385  0.109            11 TRUE       FALSE
```

All of the above, filtering by sex and taking type as a the independent variable.

Females positive correlations

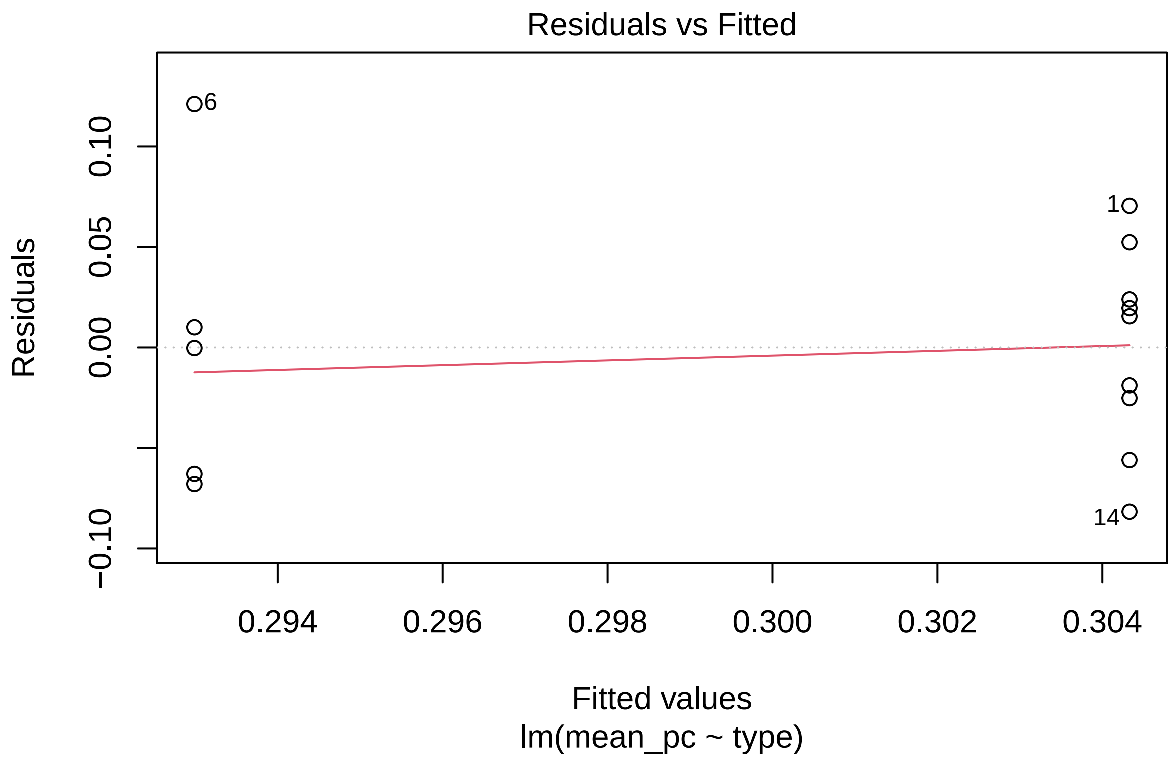
```
datos <- filter(Mean_pcorr_by_video,sex=="Female")
model  <- lm(mean_pc ~ type, data = datos)
ggqqplot(residuals(model))
```




```
shapiro_test(residuals(model))
```

```
## # A tibble: 1 x 3
##   variable         statistic p.value
##   <chr>                <dbl>   <dbl>
## 1 residuals(model)     0.962   0.750
```

```
plot(model,1)
```



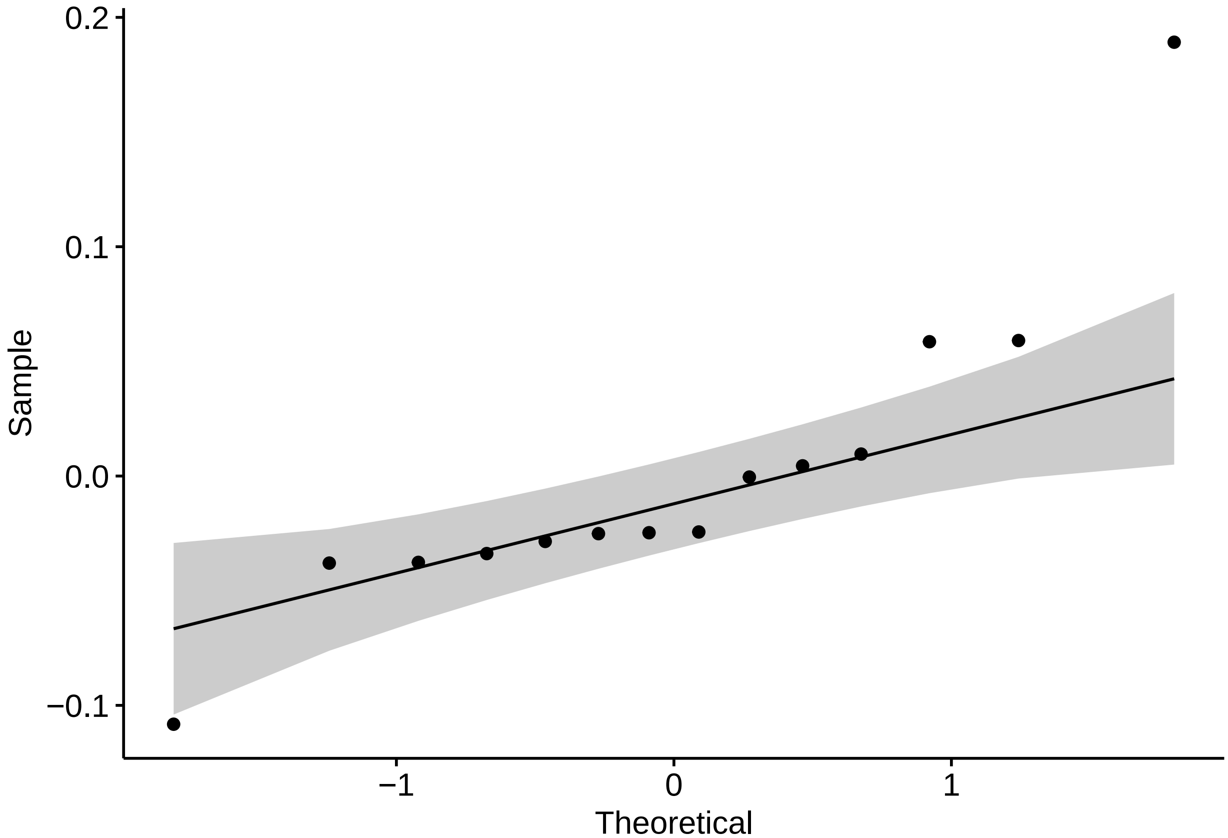

```
Mean_pcorr_by_video %>% levene_test(mean_pc ~ type)
```

```
## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.
```

```
## # A tibble: 1 x 4
##     df1   df2 statistic     p
##   <int> <int>     <dbl> <dbl>
## 1     1    17    0.0746 0.788
```

Females negative correlations

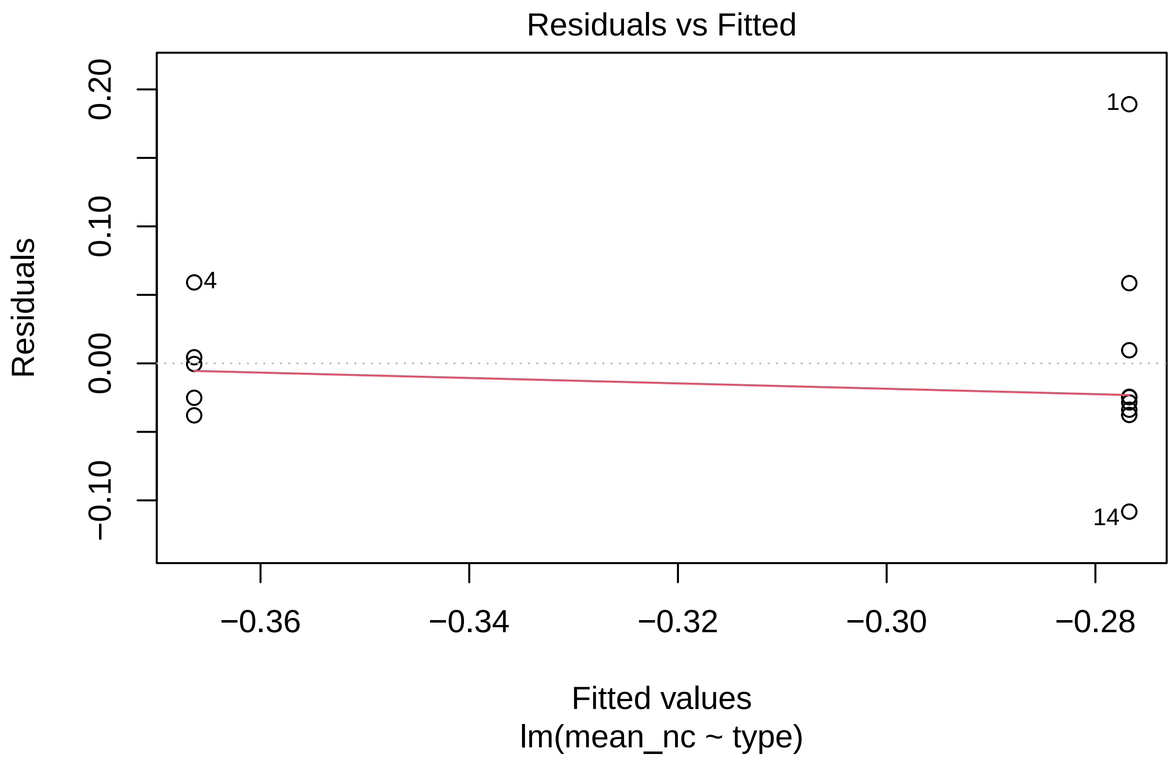
```
datos <- filter(Mean_ncorr_by_video,sex=="Female")
model  <- lm(mean_nc ~ type, data = datos)
ggqqplot(residuals(model))
```




```
shapiro_test(residuals(model))
```

```
## # A tibble: 1 x 3
##   variable         statistic p.value
##   <chr>                <dbl>   <dbl>
## 1 residuals(model)     0.833  0.0130
```

```
plot(model,1)
```

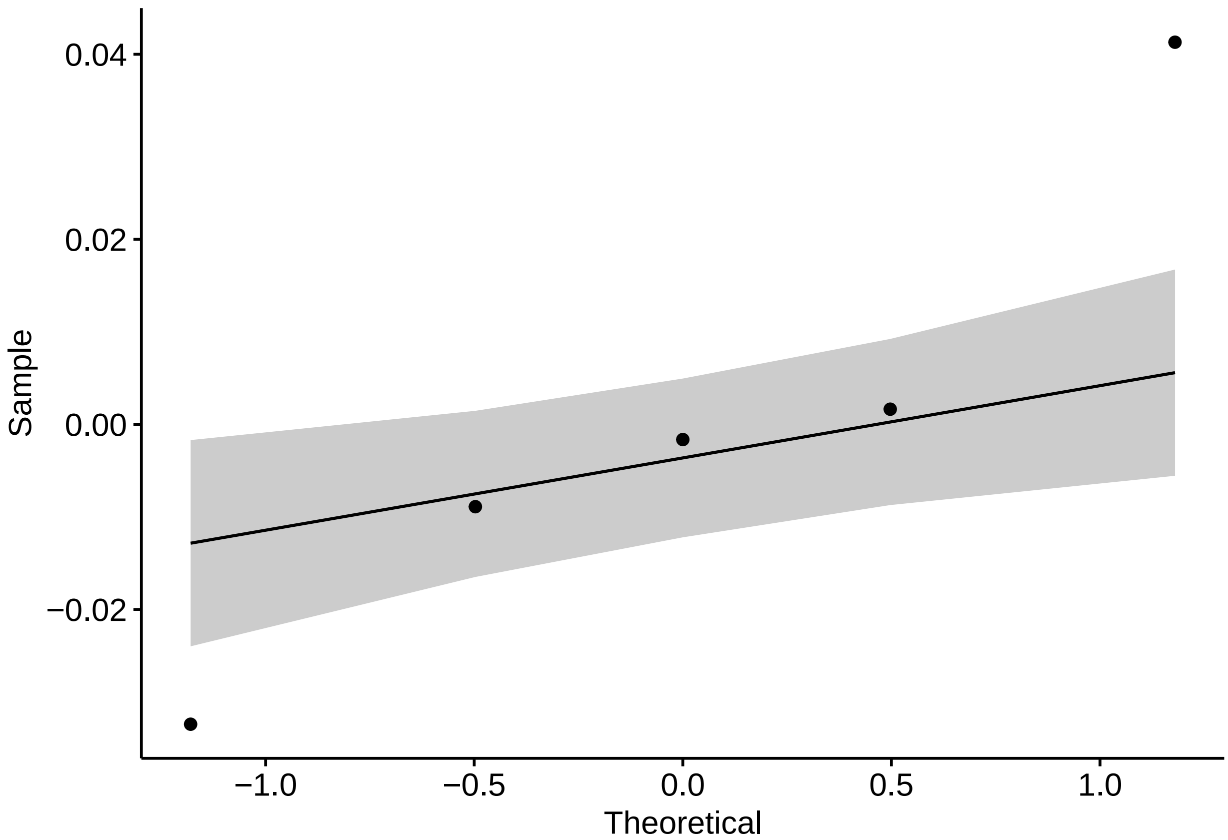



```r
Mean_ncorr_by_video %>% levene_test(mean_nc ~ type)
```

```
## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.

## # A tibble: 1 x 4
##     df1   df2 statistic     p
##   <int> <int>     <dbl> <dbl>
## 1     1    17     0.142 0.711
```

Males positive correlations

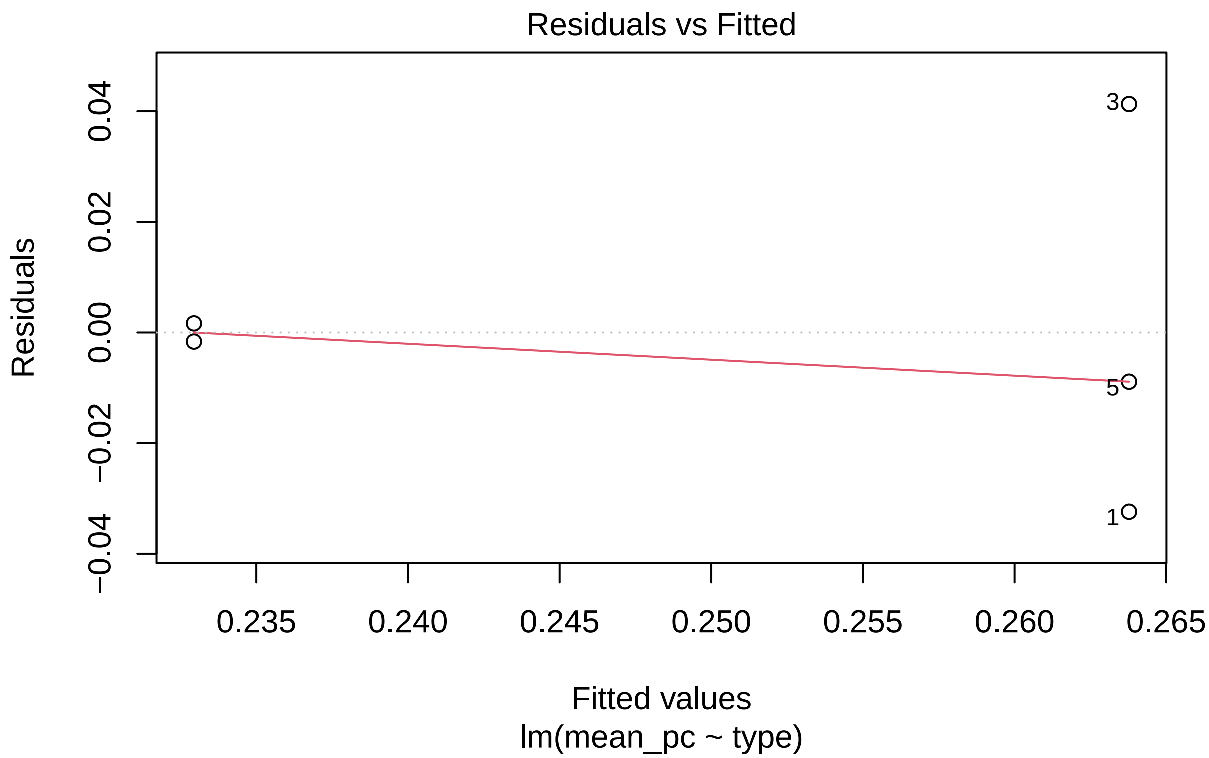
```r
datos <- filter(Mean_pcorr_by_video,sex=="Male")
model  <- lm(mean_pc ~ type, data = datos)
ggqqplot(residuals(model))
```




```
shapiro_test(residuals(model))
```

```
## # A tibble: 1 x 3
##   variable         statistic p.value
##   <chr>                <dbl>   <dbl>
## 1 residuals(model)     0.935   0.630
```

```
plot(model,1)
```



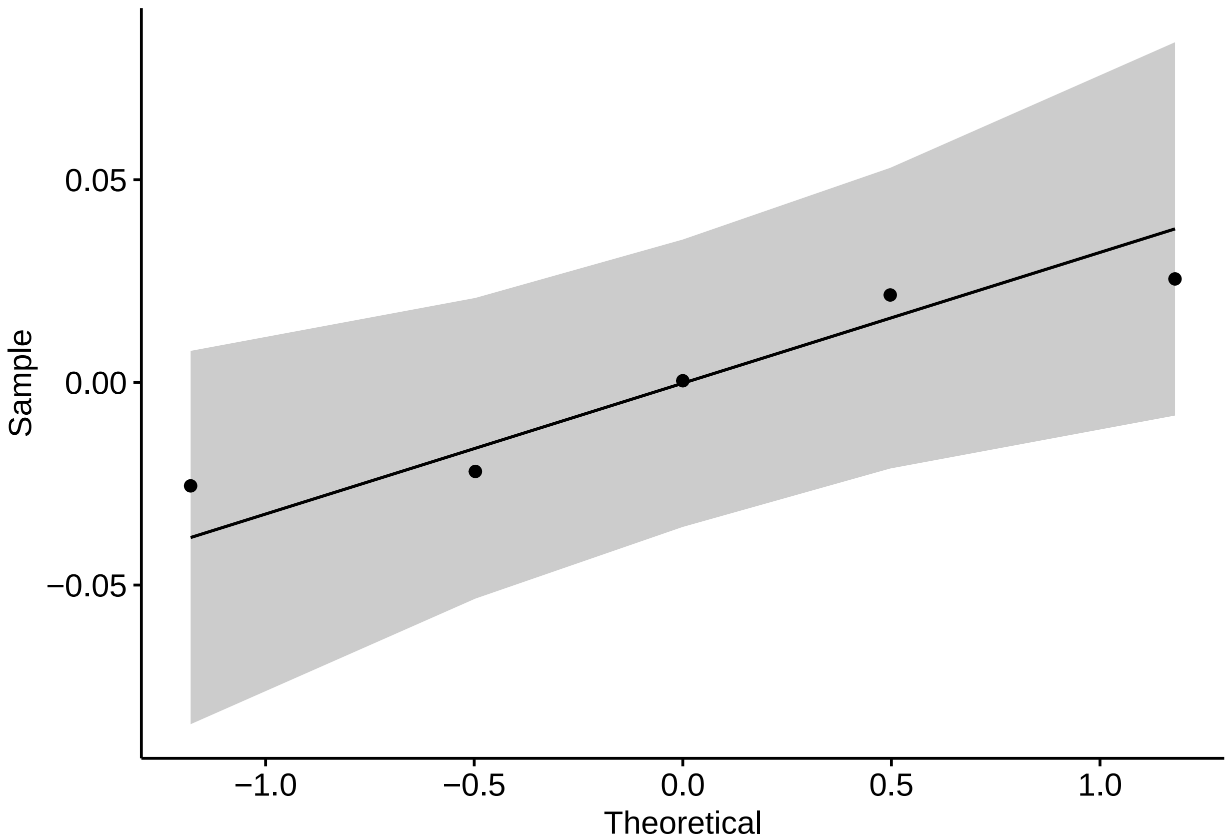

```
Mean_pcorr_by_video %>% levene_test(mean_pc ~ type)
```

```
## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.
```

```
## # A tibble: 1 x 4
##     df1   df2 statistic     p
##   <int> <int>     <dbl> <dbl>
## 1     1    17    0.0746 0.788
```

Males negative correlations

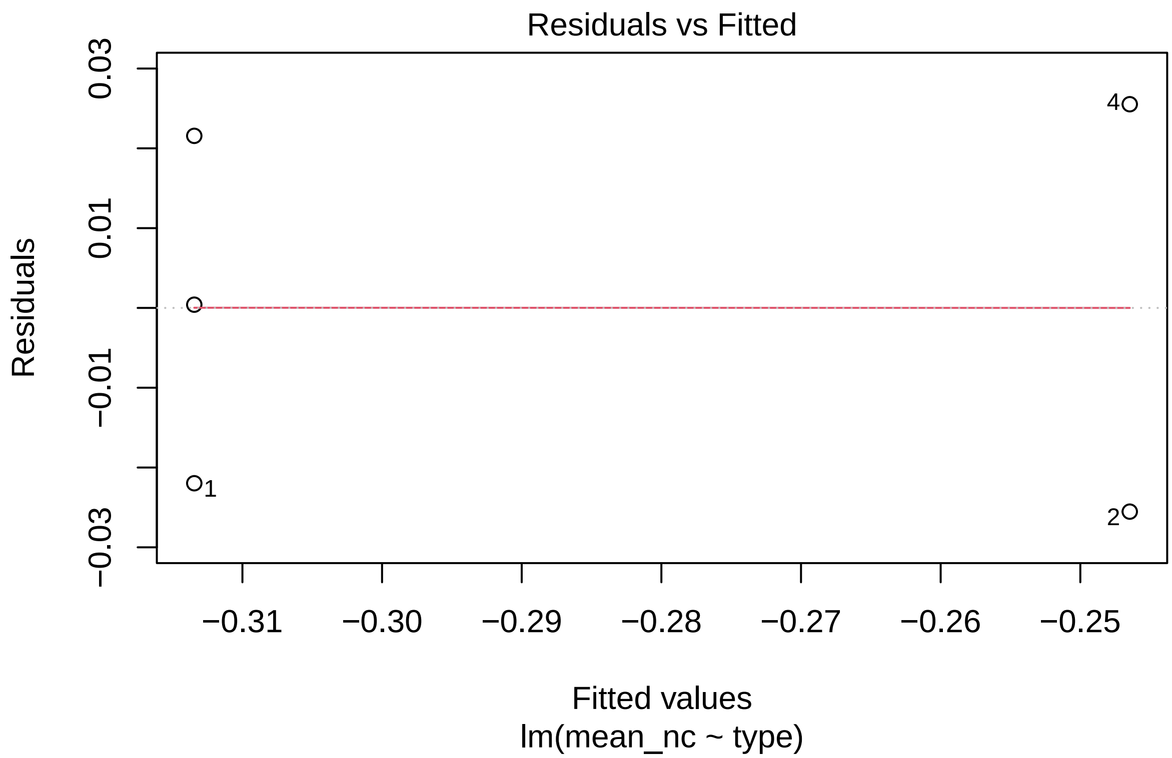
```
datos <- filter(Mean_ncorr_by_video,sex=="Male")
model  <- lm(mean_nc ~ type, data = datos)
ggqqplot(residuals(model))
```




```
shapiro_test(residuals(model))
```

```
## # A tibble: 1 x 3
##   variable         statistic p.value
##   <chr>                <dbl>   <dbl>
## 1 residuals(model)     0.877   0.297
```

```
plot(model,1)
```




```
Mean_ncorr_by_video %>% levene_test(mean_nc ~ type)
```

```
## Warning in leveneTest.default(y = y, group = group, ...): group coerced to
## factor.
```

```
## # A tibble: 1 x 4
##     df1   df2 statistic     p
##   <int> <int>     <dbl> <dbl>
## 1     1    17     0.142 0.711
```

ANOVAS

Perform one way ANOVAS, filtering for sex, and looking if type influences the *mean_pc* and *mean_nc* outcomes.

Positive correlations, male subjects

```
res.aov <- filter(Mean_pcorr_by_video,sex=='Male') %>%
  anova_test(mean_pc ~ type)
print(res.aov)
```

```
## ANOVA Table (type II tests)
##
##   Effect DFn DFd     F     p p<.05   ges
## 1   type   1   3 1.205 0.353       0.287
```

Positive correlations, female subjects

```
res.aov <- filter(Mean_pcorr_by_video,sex=='Female') %>%
  anova_test(mean_pc ~ type)
print(res.aov)
```

```
## ANOVA Table (type II tests)
##
```

```
##   Effect DFn DFd     F    p p<.05  ges
## 1   type   1  12 0.115 0.74       0.01
```

Negative correlations, male subjects

```
res.aov <- filter(Mean_ncorr_by_video,sex=='Male') %>%
  anova_test(mean_nc ~ type)
print(res.aov)
```

```
## ANOVA Table (type II tests)
##
##   Effect DFn DFd     F     p p<.05   ges
## 1   type   1   3 7.176 0.075       0.705
```

Negative correlations, male subjects (wilcox)

```
datos <- Mean_ncorr_by_video %>%
  filter(sex == "Male")

res.wcx <- wilcox.test(mean_nc ~ type, exact = TRUE, data = datos)
print(res.wcx)
```

```
##
##  Wilcoxon rank sum exact test
##
## data:  mean_nc by type
## W = 6, p-value = 0.2
## alternative hypothesis: true location shift is not equal to 0
```

Negative correlations, female subjects (wilcox)

```
datos <- Mean_ncorr_by_video %>%
  filter(sex == "Female")

res.wcx <- wilcox.test(mean_nc ~ type, exact = TRUE, data = datos)
print(res.wcx)
```

```
##
##  Wilcoxon rank sum exact test
##
## data:  mean_nc by type
## W = 5, p-value = 0.01898
## alternative hypothesis: true location shift is not equal to 0
```

Negative correlations, female subjects (anova)

```
res.aov <- filter(Mean_ncorr_by_video,sex=='Female') %>%
  anova_test(mean_nc ~ type)
print(res.aov)
```

```
## ANOVA Table (type II tests)
##
##   Effect DFn DFd     F     p p<.05   ges
## 1   type   1  12 5.062 0.044     * 0.297
```

PART 2: who leads more often patient or therapist

Upload the data and select the relevant variables for the present analysis. Add a column for the subject that leads, according to the sign of the lag. Add another column for the dominating sign of the correlation in the segment.

```
MEA_segments_leads <- read.csv("MEA_segments.csv") %>%
  transmute(
    video,
    smax     = round(spearman_max, 4),
    lag_max  = lag_spearman_max,
    smin     = round(spearman_min, 4),
    lag_min  = lag_spearman_min,
    sex,
    type
  )
x1<-which(MEA_segments_leads$smax+MEA_segments_leads$smin>0&
          abs(MEA_segments_leads$lag_max)>1)
x2<-which(MEA_segments_leads$smax+MEA_segments_leads$smin<0&
          abs(MEA_segments_leads$lag_min)>1)

MEA_segments_leads<-MEA_segments_leads[c(x1,x2), ]
MEA_segments_leads$corr_type<-"positive"
x<-which(MEA_segments_leads$smax<abs(MEA_segments_leads$smin))

MEA_segments_leads$corr_type[x]<-"negative"
MEA_segments_leads$leads<-" "
for (i in 1:nrow(MEA_segments_leads)){
    if ((MEA_segments_leads$corr_type[i]=="positive" &  MEA_segments_leads$lag_max[i]>1)
        | (MEA_segments_leads$corr_type[i]=="negative" &  MEA_segments_leads$lag_min[i]>1)){
      MEA_segments_leads$leads[i]<-"therapist"
    }
    if ((MEA_segments_leads$corr_type[i]=="positive" &  MEA_segments_leads$lag_max[i]<1)
        | (MEA_segments_leads$corr_type[i]=="negative" &  MEA_segments_leads$lag_min[i]<1)){
      MEA_segments_leads$leads[i]<-"patient"
    }
}
```

Summarize the maximal and minimal correlations.

```
quantile(MEA_segments_leads$smax,probs=seq(0,1,0.05))
```

```
##       0%       5%      10%      15%      20%      25%      30%      35% 
## -0.08930  0.05962  0.11604  0.16348  0.18532  0.20440  0.22012  0.23408 
##      40%      45%      50%      55%      60%      65%      70%      75% 
##  0.24824  0.25772  0.27190  0.28702  0.30146  0.31574  0.33016  0.35110 
##      80%      85%      90%      95%     100% 
##  0.37550  0.40128  0.43938  0.49110  0.61070
```

```
quantile(MEA_segments_leads$smin,probs=seq(0,1,0.05))
```

```
##       0%       5%      10%      15%      20%      25%      30%      35% 
## -0.69550 -0.53890 -0.47772 -0.44310 -0.40694 -0.38770 -0.36218 -0.34604 
##      40%      45%      50%      55%      60%      65%      70%      75% 
## -0.32186 -0.30974 -0.29790 -0.28652 -0.27130 -0.25974 -0.24414 -0.23000 
##      80%      85%      90%      95%     100% 
## -0.20784 -0.19120 -0.15304 -0.10826  0.02100
```

The data has a total of 360 observations.

Distribution of the segments considered. By sign of the maximal correlation:

```r
MEA_segments_leads <- group_by(MEA_segments_leads,video,sex,type,corr_type)
segments_videos_leads <- reframe(MEA_segments_leads,segments=n())
MEA_segments_leads<-ungroup(MEA_segments_leads)

segments_video_leads <- segments_videos_leads %>%      # video | corr_type | segments_pc
  pivot_wider(                                        # ← tidyr
    names_from  = corr_type,                          # columnas: positive / negative
    values_from = segments,                        # valores: recuento
    values_fill = 0                                   # si falta alguno, pon 0
  ) %>%
  arrange(video)                                      # opcional: ordena por video


segments_video_leads<-mutate(segments_video_leads,total=negative+positive)

kable(segments_video_leads)
```

video	sex	type	negative	positive	total
Vid0	Male	Treatment	12	6	18
Vid12	Female	Treatment	2	12	14
Vid13	Female	Treatment	16	9	25
Vid14	Female	Control	12	5	17
Vid15	Female	Control	9	10	19
Vid16	Female	Treatment	11	6	17
Vid17	Female	Control	6	9	15
Vid18	Male	Control	17	5	22
Vid19	Male	Treatment	9	15	24
Vid20	Female	Treatment	15	9	24
Vid21	Female	Treatment	7	12	19
Vid22	Female	Control	12	4	16
Vid23	Female	Treatment	10	5	15
Vid24	Female	Control	25	12	37
Vid25	Male	Control	7	11	18
Vid26	Female	Treatment	15	17	32
Vid27	Male	Treatment	17	8	25
Vid28	Female	Treatment	12	18	30
Vid29	Female	Treatment	12	6	18

By leading subject:

```r
MEA_segments_leads <- group_by(MEA_segments_leads,video,sex,type,leads)
segments_videos_leads <- reframe(MEA_segments_leads,segments=n())
MEA_segments_leads<-ungroup(MEA_segments_leads)

segments_video_leads <- segments_videos_leads %>%      # video | corr_type | segments_pc
  pivot_wider(                                        # ← tidyr
    names_from  = leads,                          # columnas: positive / negative
    values_from = segments,                        # valores: recuento
    values_fill = 0                                   # si falta alguno, pon 0
  ) %>%
  arrange(video)                                      # opcional: ordena por video
```

```
segments_video_leads<-mutate(segments_video_leads,total=therapist+patient,ratio=(patient-therapist)/tota

kable(segments_video_leads)
```

video	sex	type	patient	therapist	total	ratio
Vid0	Male	Treatment	10	8	18	0.1111111
Vid12	Female	Treatment	5	9	14	-0.2857143
Vid13	Female	Treatment	10	15	25	-0.2000000
Vid14	Female	Control	9	8	17	0.0588235
Vid15	Female	Control	9	10	19	-0.0526316
Vid16	Female	Treatment	8	9	17	-0.0588235
Vid17	Female	Control	3	12	15	-0.6000000
Vid18	Male	Control	9	13	22	-0.1818182
Vid19	Male	Treatment	15	9	24	0.2500000
Vid20	Female	Treatment	13	11	24	0.0833333
Vid21	Female	Treatment	11	8	19	0.1578947
Vid22	Female	Control	5	11	16	-0.3750000
Vid23	Female	Treatment	7	8	15	-0.0666667
Vid24	Female	Control	14	23	37	-0.2432432
Vid25	Male	Control	12	6	18	0.3333333
Vid26	Female	Treatment	6	26	32	-0.6250000
Vid27	Male	Treatment	13	12	25	0.0400000
Vid28	Female	Treatment	15	15	30	0.0000000
Vid29	Female	Treatment	8	10	18	-0.1111111

By leading subject, only positive correlations:

```
MEA_segments_leads_pos<-filter(MEA_segments_leads,smax>0.25)
MEA_psegments_leads <- group_by(filter(MEA_segments_leads,corr_type=="positive"),
                                video,sex,type,leads)
psegments_videos_leads <- reframe(MEA_psegments_leads,segments=n())

psegments_video_leads <- psegments_videos_leads %>%
  pivot_wider(                                          # ← tidyr
    names_from  = leads,                              # columnas: positive / negative
    values_from = segments,                            # valores: recuento
    values_fill = 0                                     # si falta alguno, pon 0
  ) %>%
  arrange(video)                                        # opcional: ordena por video


psegments_video_leads<-mutate(psegments_video_leads,total=therapist+patient,ratio=(patient-therapist)/t

kable(psegments_video_leads)
```

video	sex	type	patient	therapist	total	ratio
Vid0	Male	Treatment	3	3	6	0.0000000
Vid12	Female	Treatment	3	9	12	-0.5000000
Vid13	Female	Treatment	3	6	9	-0.3333333
Vid14	Female	Control	4	1	5	0.6000000
Vid15	Female	Control	5	5	10	0.0000000

video	sex	type	patient	therapist	total	ratio
Vid16	Female	Treatment	2	4	6	-0.3333333
Vid17	Female	Control	3	6	9	-0.3333333
Vid18	Male	Control	0	5	5	-1.0000000
Vid19	Male	Treatment	9	6	15	0.2000000
Vid20	Female	Treatment	4	5	9	-0.1111111
Vid21	Female	Treatment	7	5	12	0.1666667
Vid22	Female	Control	3	1	4	0.5000000
Vid23	Female	Treatment	3	2	5	0.2000000
Vid24	Female	Control	6	6	12	0.0000000
Vid25	Male	Control	6	5	11	0.0909091
Vid26	Female	Treatment	6	11	17	-0.2941176
Vid27	Male	Treatment	5	3	8	0.2500000
Vid28	Female	Treatment	8	10	18	-0.1111111
Vid29	Female	Treatment	5	1	6	0.6666667

```
write.csv(psegments_video_leads,"Table_N4.csv",row.names = FALSE)
```

By leading subject, only negative correlations:

```
MEA_segments_leads_neg<-filter(MEA_segments_leads,smin< -0.25)
MEA_nsegments_leads <- group_by(filter(MEA_segments_leads_neg,corr_type=="negative"),
                                video,sex,type,leads)
nsegments_videos_leads <- reframe(MEA_nsegments_leads,segments=n())

nsegments_video_leads <- nsegments_videos_leads %>%
  pivot_wider(                                          # ← tidyr
    names_from  = leads,                            # columnas: positive / negative
    values_from = segments,                          # valores: recuento
    values_fill = 0                                     # si falta alguno, pon 0
  ) %>%
  arrange(video)                                        # opcional: ordena por video


nsegments_video_leads<-mutate(nsegments_video_leads,total=therapist+patient,ratio=(patient-therapist)/t

kable(nsegments_video_leads)
```

video	sex	type	patient	therapist	total	ratio
Vid0	Male	Treatment	6	5	11	0.0909091
Vid12	Female	Treatment	2	0	2	1.0000000
Vid13	Female	Treatment	6	9	15	-0.2000000
Vid14	Female	Control	4	6	10	-0.2000000
Vid15	Female	Control	3	5	8	-0.2500000
Vid16	Female	Treatment	5	5	10	0.0000000
Vid17	Female	Control	0	6	6	-1.0000000
Vid18	Male	Control	9	5	14	0.2857143
Vid19	Male	Treatment	6	3	9	0.3333333
Vid20	Female	Treatment	9	6	15	0.2000000
Vid21	Female	Treatment	4	3	7	0.1428571
Vid22	Female	Control	1	10	11	-0.8181818
Vid23	Female	Treatment	4	5	9	-0.1111111

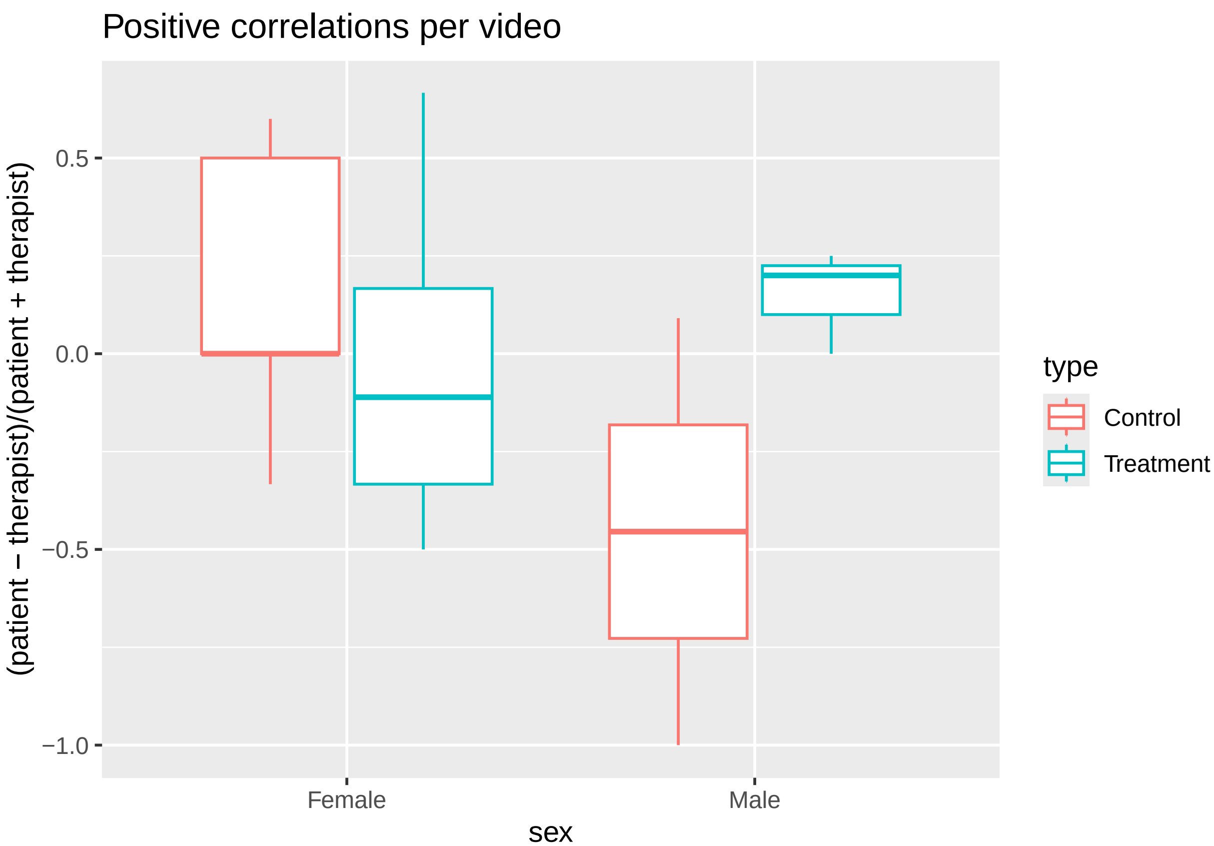
video	sex	type	patient	therapist	total	ratio
Vid24	Female	Control	7	14	21	-0.3333333
Vid25	Male	Control	5	1	6	0.6666667
Vid26	Female	Treatment	0	14	14	-1.0000000
Vid27	Male	Treatment	8	9	17	-0.0588235
Vid28	Female	Treatment	7	5	12	0.1666667
Vid29	Female	Treatment	3	9	12	-0.5000000

```
write.csv(nsegments_video_leads,"Table_N5.csv",row.names = FALSE)
```

Boxplot visualization.

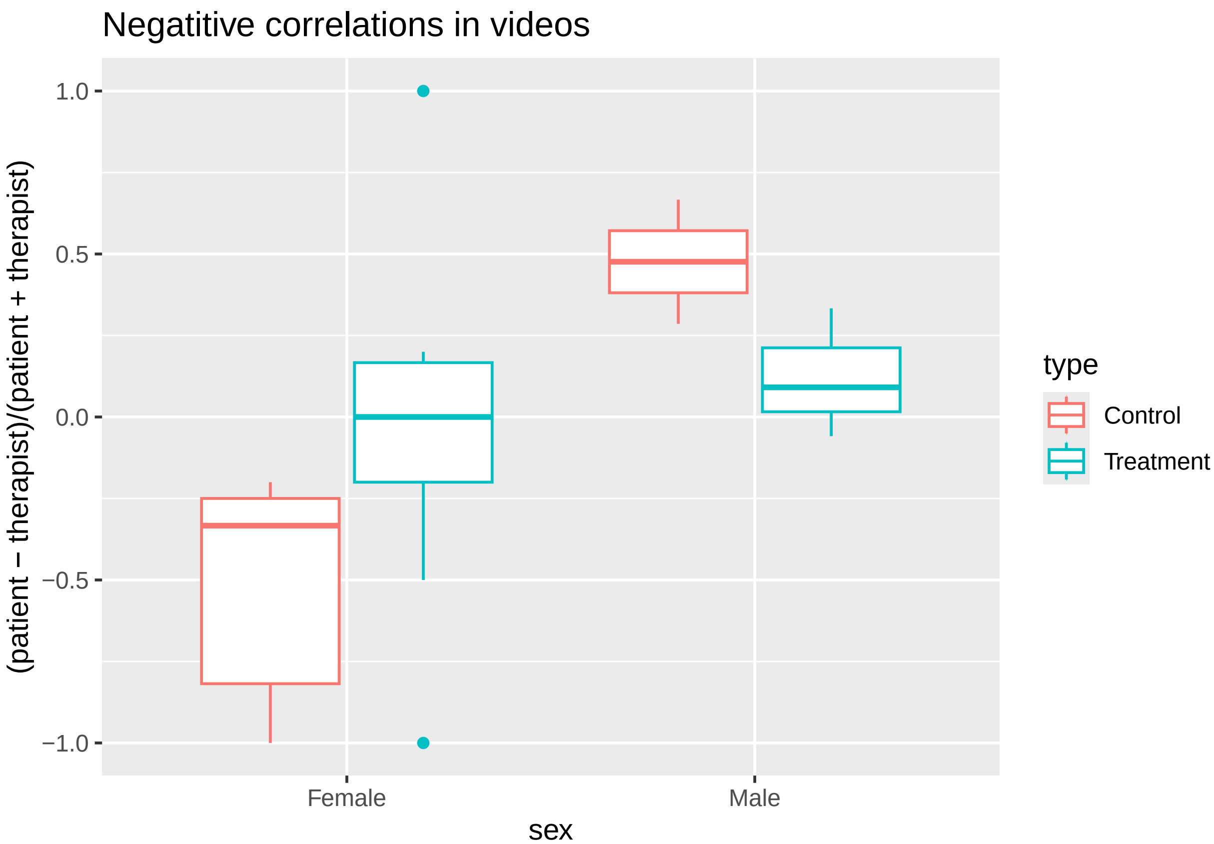
Positive correlations

```
p1<-ggplot(psegments_video_leads,aes(x=sex,y=(patient-therapist)/(patient+therapist),color=type))+geom_l
show(p1)
```



Negative correlations

```
p2<-ggplot(nsegments_video_leads,aes(x=sex,y=(patient-therapist)/(patient+therapist),color=type))+geom_l
show(p2)
```



Statistical analysis for PART 2.

Anovas for positive correlations, controlled by gender

```
datos<-psegments_video_leads
anova <- filter(datos,sex=='Female') %>% anova_test(ratio ~ type)
print(anova)
```

```
## ANOVA Table (type II tests)
##
##   Effect DFn DFd     F     p p<.05  ges
## 1   type   1  12 1.187 0.297       0.09
```

```
anova <- filter(datos,sex=='Male') %>% anova_test(ratio ~ type)
print(anova)
```

```
## ANOVA Table (type II tests)
##
##   Effect DFn DFd     F     p p<.05  ges
## 1   type   1   3 2.088 0.244       0.41
```

Anovas for negative correlations, controlled by gender

```
datos<-nsegments_video_leads
anova <- filter(datos,sex=='Female') %>% anova_test(ratio ~ type)
print(anova)
```

```
## ANOVA Table (type II tests)
##
```

```
##   Effect DFn DFd     F     p p<.05   ges
## 1   type   1  12 3.141 0.102       0.207
```

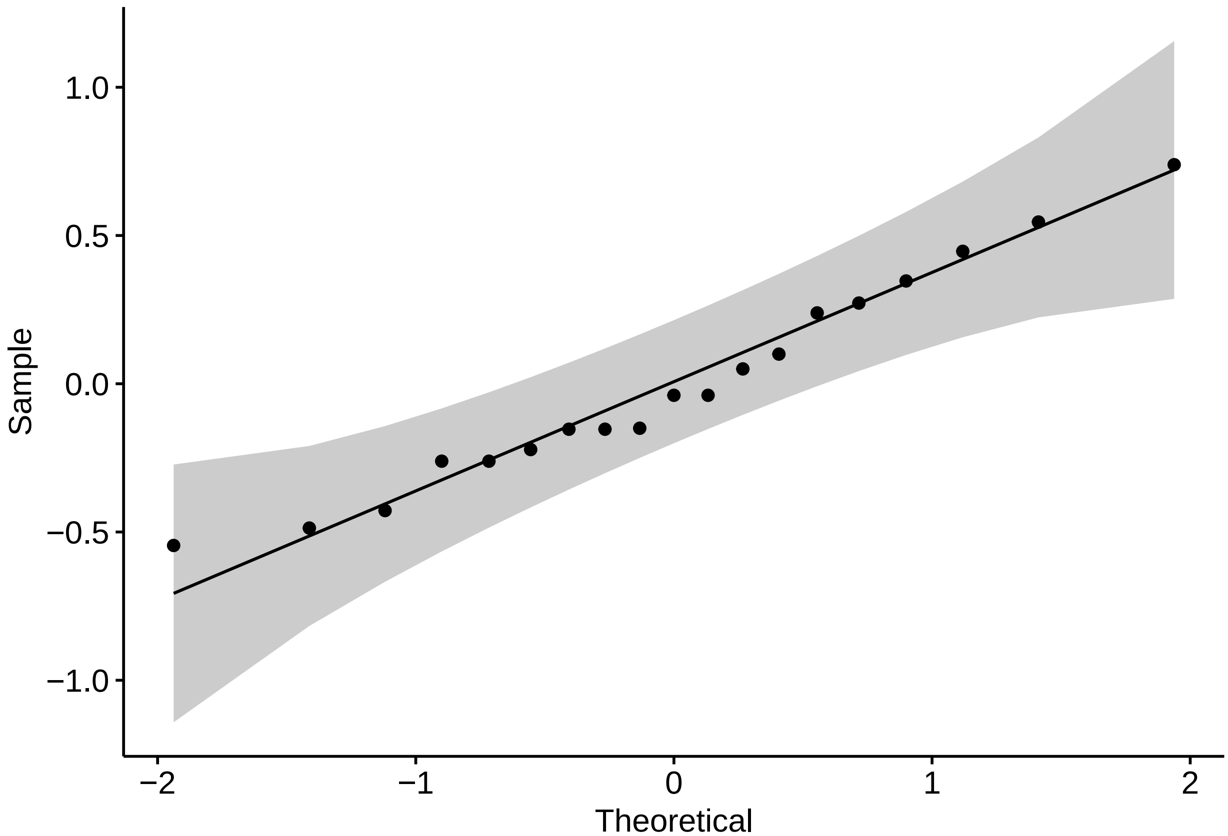
```
anova <- filter(datos,sex=='Male') %>% anova_test(ratio ~ type)
print(anova)
```

```
## ANOVA Table (type II tests)
##
##   Effect DFn DFd     F     p p<.05 ges
## 1   type   1   3 2.996 0.182       0.5
```

Shapiro and Levene tests

Positive correlations

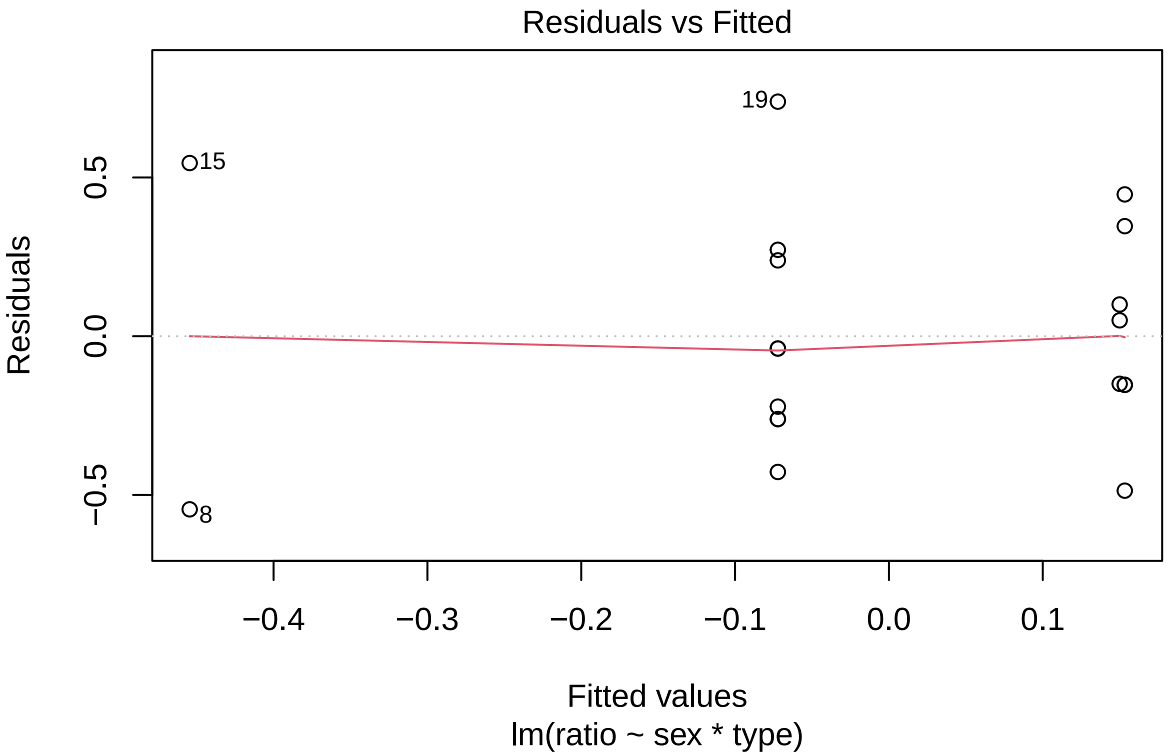
```
datos<-psegments_video_leads #positive correlations
model  <- lm(ratio ~ sex*type, data = datos)
ggqqplot(residuals(model))
```




```
shapiro_test(residuals(model))
```

```
## # A tibble: 1 x 3
##   variable         statistic p.value
##   <chr>                <dbl>   <dbl>
## 1 residuals(model)     0.967   0.715
```

```r
plot(model,1)
```

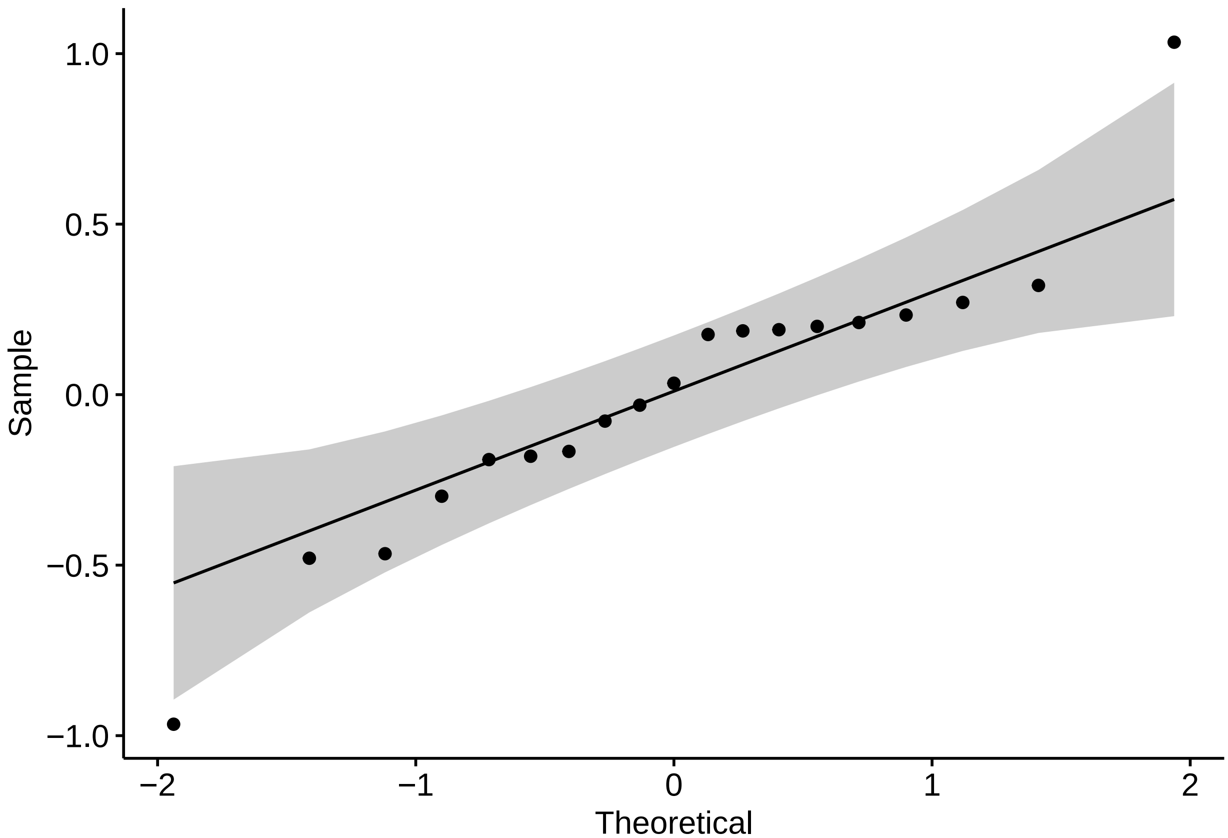



```r
datos %>% levene_test(ratio ~ sex*type)#positive correlations
```

```
## # A tibble: 1 x 4
##     df1   df2 statistic     p
##   <int> <int>     <dbl> <dbl>
## 1     3    15      1.69 0.212
```

Negative correlations

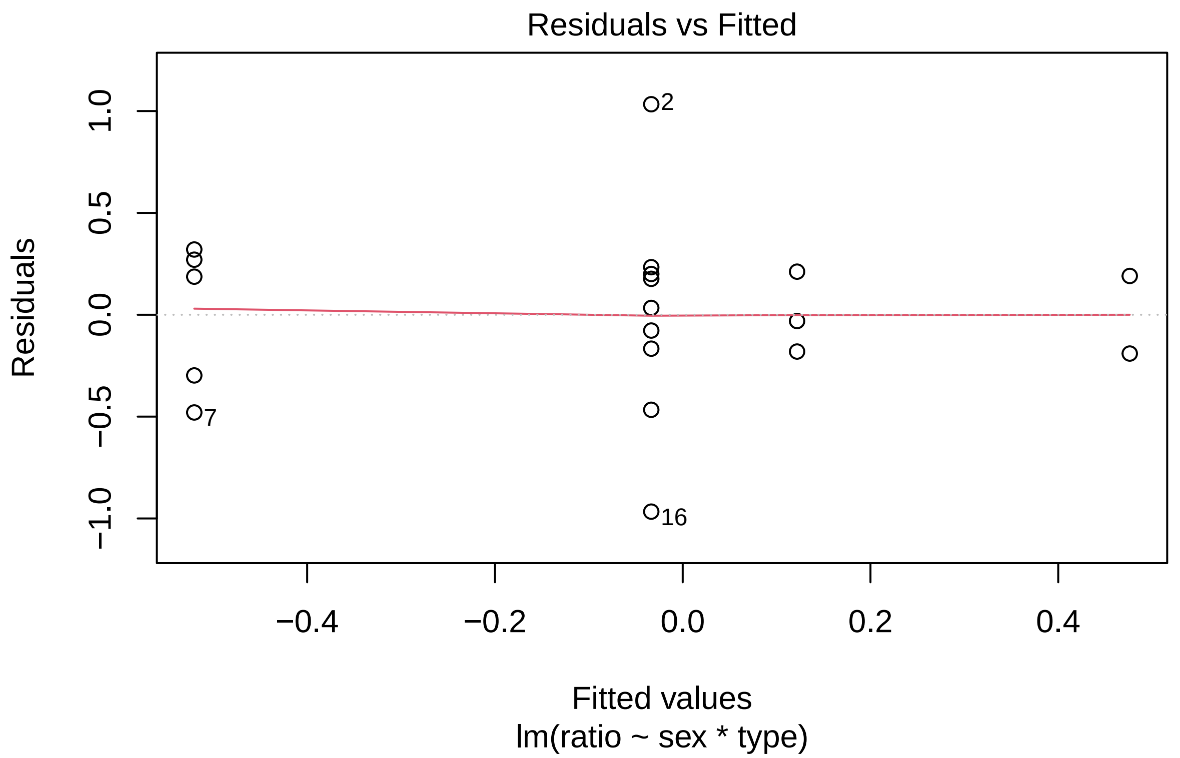
```r
datos<-nsegments_video_leads #negative correlations
model  <- lm(ratio ~ sex*type, data = datos)
ggqqplot(residuals(model))
```




```
shapiro_test(residuals(model))
```

```
## # A tibble: 1 x 3
##   variable         statistic p.value
##   <chr>                <dbl>   <dbl>
## 1 residuals(model)     0.931   0.182
```

```
plot(model,1)
```




```
datos %>% levene_test(ratio ~ sex*type) #negative correlations
```

```
## # A tibble: 1 x 4
##     df1   df2 statistic     p
##   <int> <int>     <dbl> <dbl>
## 1     3    15     0.502 0.686
```

Outliers. Positive correlations.

```
datos<-psegments_video_leads
temp<-datos%>%group_by(sex,type)%>%identify_outliers(ratio)
print(temp)
```

```
## [1] sex        type       video      patient    therapist  total      ratio
## [8] is.outlier is.extreme
## <0 rows> (o 0- extensión row.names)
```

Outliers. Negative correlations.

```
datos<-nsegments_video_leads
temp<-datos%>%group_by(sex,type)%>%identify_outliers(ratio)
print(temp)
```

```
## # A tibble: 2 x 9
##   sex    type      video patient therapist total ratio is.outlier is.extreme
##   <chr>  <chr>     <chr>   <int>     <int> <int> <dbl> <lgl>      <lgl>
## 1 Female Treatment Vid12       2         0     2     1 TRUE       FALSE
## 2 Female Treatment Vid26       0        14    14    -1 TRUE       FALSE
```

PART 3: does the diagnosis affect the sign of the correlation

Here, we look on whether the therapist movement is mirrored (positive correlation) or inhibited (negative correlation) by the patient. We start uploading and filtering the data to keep only the segments for which the therapist leads.

```
MEA_segments_sign <- read.csv("MEA_segments.csv") %>%
  transmute(
    video,
    smax     = round(spearman_max, 4),
    lag_max  = lag_spearman_max,
    smin     = round(spearman_min, 4),
    lag_min  = lag_spearman_min,
    sex,
    type
  )
x1<-which(MEA_segments_sign$smax+MEA_segments_sign$smin>0&
            abs(MEA_segments_sign$lag_max)>1)
x2<-which(MEA_segments_sign$smax+MEA_segments_sign$smin<0&
            abs(MEA_segments_sign$lag_min)>1)

MEA_segments_sign<-MEA_segments_sign[c(x1,x2), ] #cut when lags of extremal correlations is less than 1


MEA_segments_sign$corr_type<-"positive"
x<-which(MEA_segments_sign$smax<abs(MEA_segments_sign$smin))
MEA_segments_sign$corr_type[x]<-"negative"


MEA_segments_sign$leads<-" "
for (i in 1:nrow(MEA_segments_sign)){
  if ((MEA_segments_sign$corr_type[i]=="positive" &  MEA_segments_sign$lag_max[i]>1)
      | (MEA_segments_sign$corr_type[i]=="negative" &  MEA_segments_sign$lag_min[i]>1)){
    MEA_segments_sign$leads[i]<-"therapist"
  }
  if ((MEA_segments_sign$corr_type[i]=="positive" &  MEA_segments_sign$lag_max[i]<1)
      | (MEA_segments_sign$corr_type[i]=="negative" &  MEA_segments_sign$lag_min[i]<1)){
    MEA_segments_sign$leads[i]<-"patient"
  }
}

MEA_leads_tp<-filter(MEA_segments_sign,leads=="therapist")%>%
  filter(max(smax,abs(smin))>0.15)%>% # Only segments where therapist leads
  select(-leads)
```

The resulting database *MEA_leads_tp* consists of 223 observations (1 minute segments). Next, for each video we count the number of segments with positive and negative dominant correlations. We define the variable *mean_corr* as follows: for each segment observed, take the sum of the maximal and minimal correlations, and take the average of this results across all the segments for each video.

```
MEA_leads_tp <- group_by(MEA_leads_tp,video,sex,type,corr_type)
segments_corr_sign <- reframe(MEA_leads_tp,segments=n())
MEA_leads_tp<-ungroup(MEA_leads_tp)
segments_corr_sign <- segments_corr_sign  %>%      # create table with number of positive/negative by v
   pivot_wider(
     names_from  = corr_type,
     values_from = segments,
     values_fill = 0
   ) %>%
```

```
  arrange(video)%>%
  mutate(total=negative+positive,ratio=(positive-negative)/total)

MEA_leads_tp<-group_by(MEA_leads_tp,video,sex,type)
MEA_corr_sign<-reframe(MEA_leads_tp,mean_corr=sum(smax+smin)/n())%>%
  select(video,mean_corr)
MEA_leads_tp<-ungroup(MEA_leads_tp)

MEA_corr_sign<-merge(segments_corr_sign,MEA_corr_sign,by="video")%>%
 select(-ratio)
MEA_corr_sign$type<-as.factor(MEA_corr_sign$type)
```

```
kable(MEA_corr_sign)
```

video	sex	type	negative	positive	total	mean_corr
Vid0	Male	Treatment	5	3	8	-0.0971500
Vid12	Female	Treatment	0	9	9	0.1860000
Vid13	Female	Treatment	9	6	15	-0.0284000
Vid14	Female	Control	7	1	8	-0.2417500
Vid15	Female	Control	5	5	10	-0.0740300
Vid16	Female	Treatment	5	4	9	-0.0188222
Vid17	Female	Control	6	6	12	0.0132583
Vid18	Male	Control	8	5	13	-0.0446692
Vid19	Male	Treatment	3	6	9	0.0447556
Vid20	Female	Treatment	6	5	11	0.0483636
Vid21	Female	Treatment	3	5	8	0.0120750
Vid22	Female	Control	10	1	11	-0.2798909
Vid23	Female	Treatment	6	2	8	-0.0000750
Vid24	Female	Control	17	6	23	-0.1109522
Vid25	Male	Control	1	5	6	0.1615167
Vid26	Female	Treatment	15	11	26	-0.0220269
Vid27	Male	Treatment	9	3	12	-0.0739917
Vid28	Female	Treatment	5	10	15	0.0279600
Vid29	Female	Treatment	9	1	10	-0.1691400

Frequency of negative and positive correlations for each group

```
MEA_leads_tp<-group_by(MEA_leads_tp,sex,type,corr_type)
MEA_which_sign<-reframe(MEA_leads_tp,segments=n())
MEA_leads_tp<-ungroup(MEA_leads_tp)

kable(MEA_which_sign)
```

sex	type	corr_type	segments
Female	Control	negative	45
Female	Control	positive	19
Female	Treatment	negative	58
Female	Treatment	positive	53
Male	Control	negative	9
Male	Control	positive	10
Male	Treatment	negative	17
Male	Treatment	positive	12

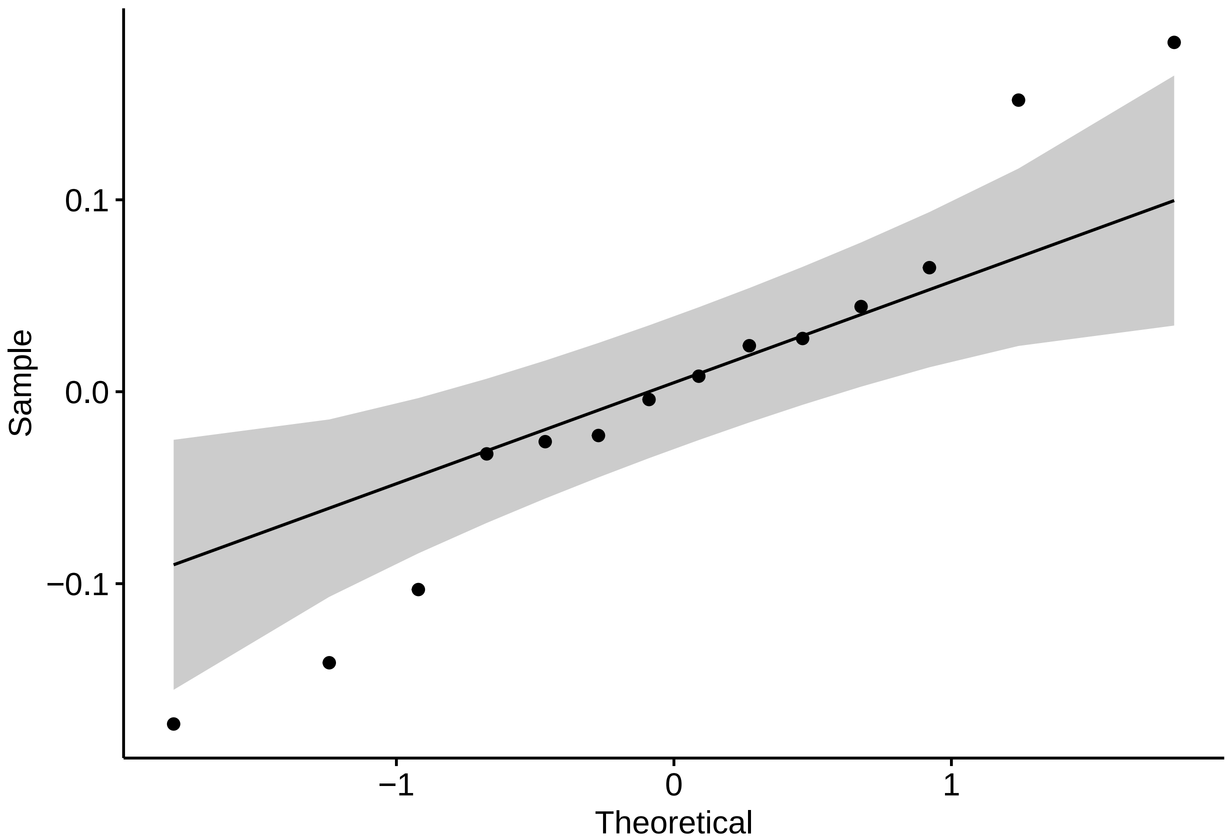
Statistical analysis for PART 3.

Separate lists by gender

```
MEA_fsign_freq<-filter(MEA_which_sign,sex=="Female")%>%
  select(-sex)

MEA_msign_freq<-filter(MEA_which_sign,sex=="Male")%>%
  select(-sex)


MEA_fsign<-filter(MEA_corr_sign,sex=="Female")%>%
  select(-sex)

MEA_msign<-filter(MEA_corr_sign,sex=="Male")%>%
  select(-sex)
```

Shapiro and Levene tests

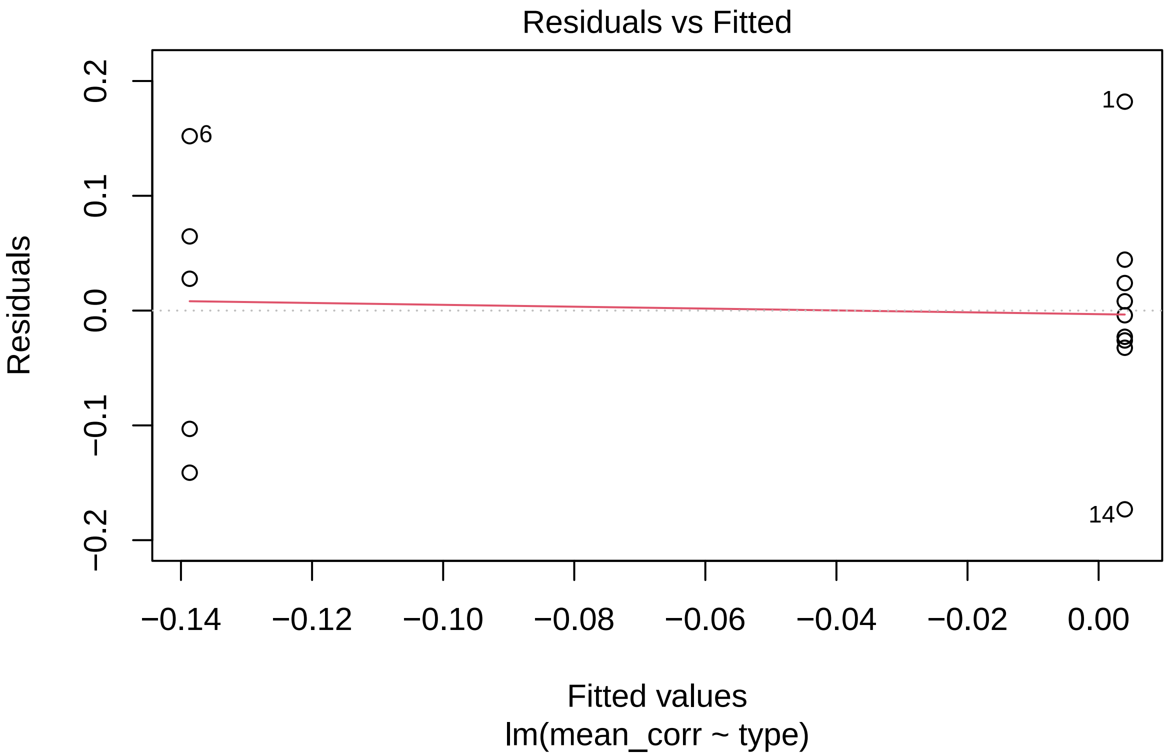
```
datos<-MEA_fsign  #shapiro and levene tests for female groups
model  <- lm(mean_corr ~ type, data = datos)
ggqqplot(residuals(model))
```




```
print(shapiro_test(residuals(model)))
```

```
## # A tibble: 1 x 3
##   variable         statistic p.value
##   <chr>                <dbl>   <dbl>
## 1 residuals(model)     0.963   0.777
```

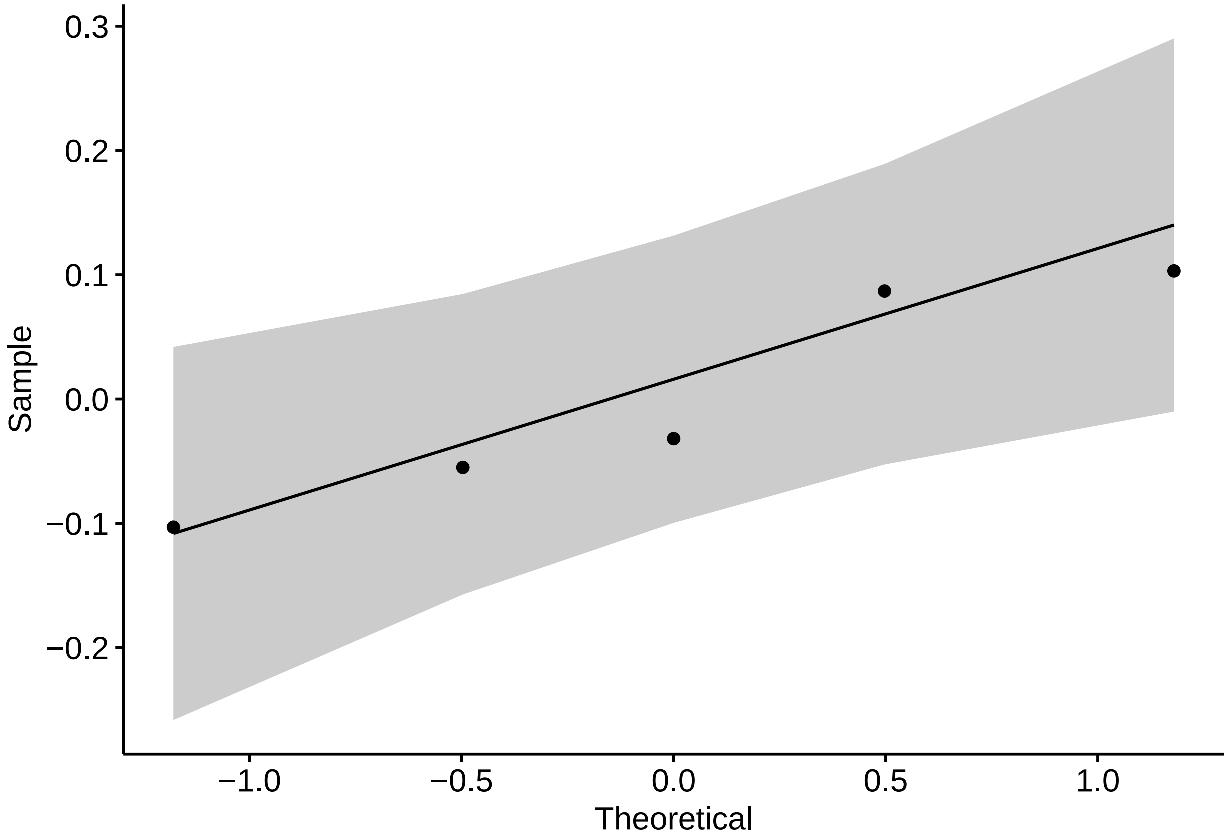
```r
plot(model,1)
```




```r
lev<-datos %>% levene_test(mean_corr ~ type)
print(c("levene females",lev))
```

```
## [[1]]
## [1] "levene females"
##
## $df1
## [1] 1
##
## $df2
## [1] 12
##
## $statistic
## [1] 0.81176
##
## $p
## [1] 0.3853262
```

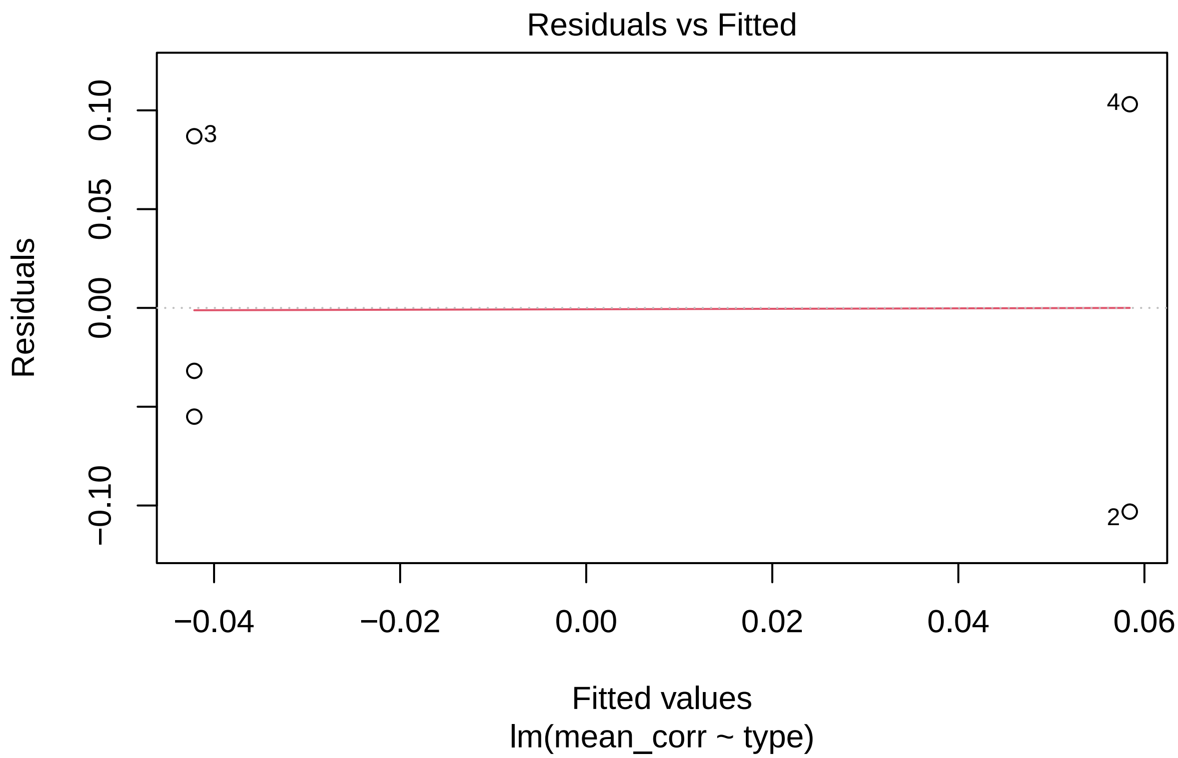
```r
datos<-MEA_msign  #shapiro and levene tests for male groups
model  <- lm(mean_corr ~ type, data = datos)
ggqqplot(residuals(model))
```




```r
print(shapiro_test(residuals(model)))
```

```
## # A tibble: 1 x 3
##   variable          statistic p.value
##   <chr>                 <dbl>   <dbl>
## 1 residuals(model)      0.893   0.374
```

```r
plot(model,1)
```




```
lev<-datos %>% levene_test(mean_corr ~ type)
print(c("levene males",lev))
```

```
## [[1]]
## [1] "levene males"
##
## $df1
## [1] 1
##
## $df2
## [1] 3
##
## $statistic
## [1] 1.41398
##
## $p
## [1] 0.3199388
```

Anovas for mean_corr

```
anova <- MEA_fsign %>% anova_test(mean_corr ~  type) #female cohort

print(anova)
```

```
## ANOVA Table (type II tests)
##
##   Effect DFn DFd     F     p p<.05   ges
## 1   type   1  12 6.203 0.028     * 0.341
```

```
anova <- MEA_msign %>% anova_test(mean_corr ~  type) #male cohort

print(anova)
```

```
## ANOVA Table (type II tests)
##
##   Effect DFn DFd     F    p p<.05  ges
## 1   type   1   3 1.108 0.37       0.27
```